ELEC3104: Digital Signal Processing

Topic 2: Discrete-Time Systems (pp. 2-5)

- Classification of Discrete-Time Systems (pp. 6-8)
- Linear Time Invariant (LTI) Systems (pp. 9-14)
- Introduction to z-Transform (pp. 15-23)
- Pole-Zero Description of Discrete-Time Systems (pp. 24-28)
- Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT) (pp. 29-42)
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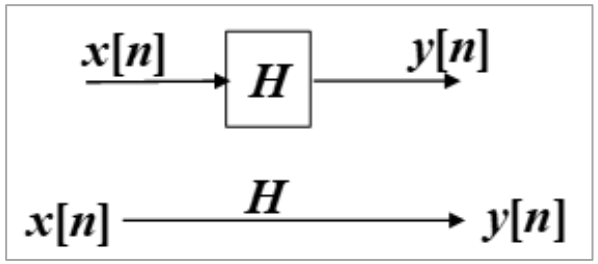
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Discrete-time systems:

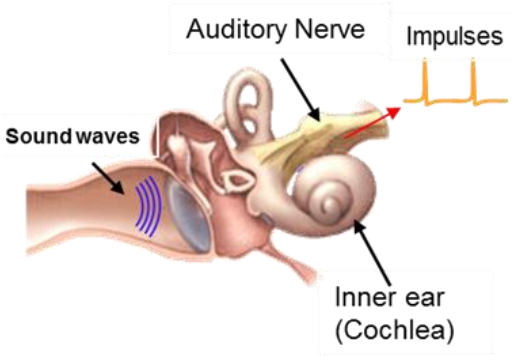
Introduction

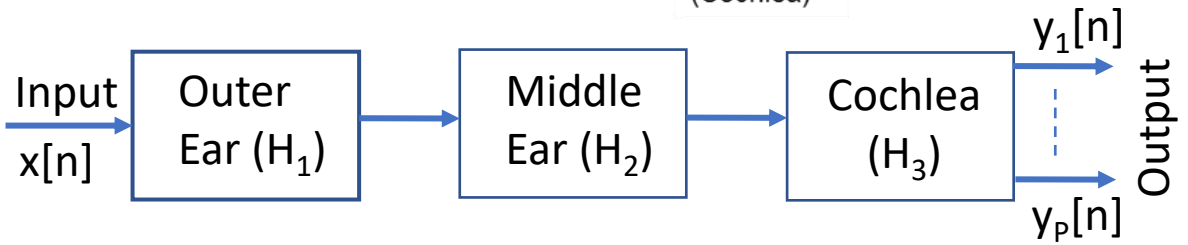
- A discrete-time system, *H*, is a device (or an algorithm) that operates on a discrete-time signal called the input *x*[*n*] (or excitation) according to some well defined rule, to produce another discrete-time signal, *y*[*n*], called the output (or response) – see diagram below



- The input signal *x*[*n*] is transformed by the system *H* to produce a signal *y*[*n*], is expressed by:
$y[n] = H\{x[n]\}$

Peripheral auditory system





For example, it is possible to implement a discrete time model of the peripheral auditory system comprising of a model of the outer ear (H_1) and the middle ear (H_2) and the model of the cochlea (H_3).

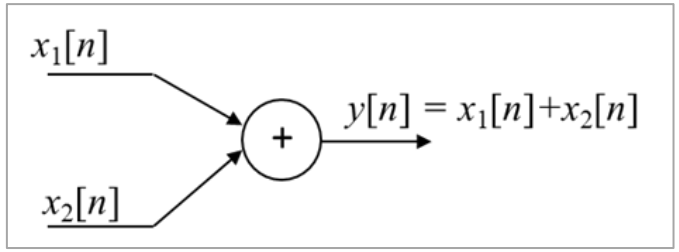
Discrete-time systems:

Block Diagram Representation

- In order to introduce a block diagram representation of discrete-time systems, we need to define some basic blocks that can be interconnected to form complex systems.

An adder

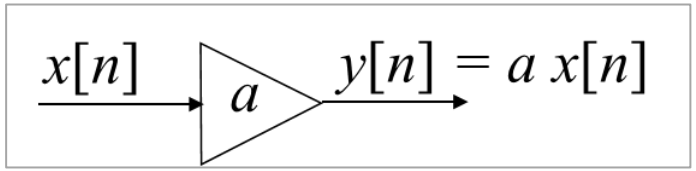
- A system that performs the addition of two signal sequences (x_1[n] & x_2[n]) to form another sequence $y[n]$.



- Note that it is not necessary to store either one of the sequences in order to perform the addition. In other words, the addition operation is **memoryless**.

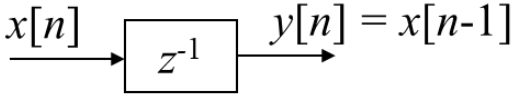
A constant multiplier

- This operation simply represents applying a scale factor 'a' on the input $x[n]$. Note that this operation is also **memoryless**.



A unit delay element

- The unit delay is a special system that simply delays the signal passing through it by one sample (or sample period).
- If the input signal is $x[n]$, the output is $x[n-1]$. In fact, the sample $x[n-1]$ is stored in memory at time $n-1$ and it is recalled from memory at time n to form $y[n] = x[n-1]$.
- Thus this basic building block **requires memory**. We use the symbol T or z^{-1} to denote the unit of delay.



n	-1	0	1	2	3
x[n]	x[-1]=0	x[0]=1	x[1]=0	x[2]=5	x[3]=7
y[n]	Y[-1]=?	Y[0]=0	Y[1]=1	Y[2]=0	Y[3]=5

Discrete-time systems:

Difference Equations

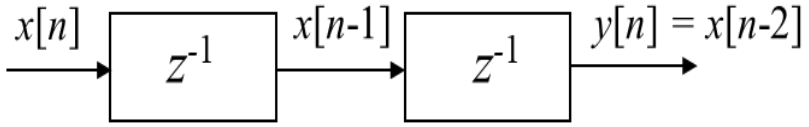
- A discrete-time system consisting of combinations of adders, multipliers and unit delays can always be described by a set of **difference equations**.
- The equations would be ordinary algebraic equations if no delays were present.
- For example, the system described by:

$$y[n] = a_0 x[n] + b_1 y[n-1]$$

produces the present output $y[n]$ as the weighted sum of the previous output $y[n-1]$ and the present input $x[n]$. The quantities a_0 and b_1 are called *coefficients*.

Example 1:

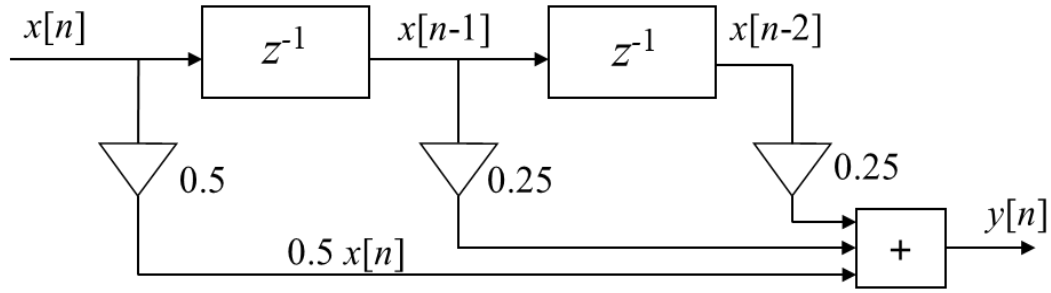
- Draw a system implementation for the difference equation: $y[n] = x[n-2]$



Example 2:

- Draw a system implementation for the difference equation:

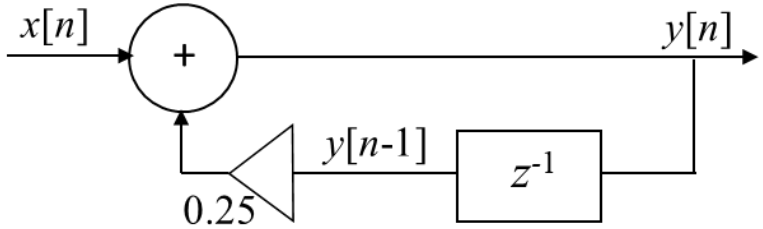
$$y[n] = \frac{1}{2}x[n] + \frac{1}{4}x[n-1] + \frac{1}{4}x[n-2]$$



Example 3:

- Draw a system implementation for the difference equation:

$$y[n] = x[n] + 0.25y[n-1]$$



Discrete-time systems

Example 4:

- Draw a system implementation for the difference equation: $y[n] = a_0 x[n] + a_1 x[n-1] - b_1 y[n-1]$

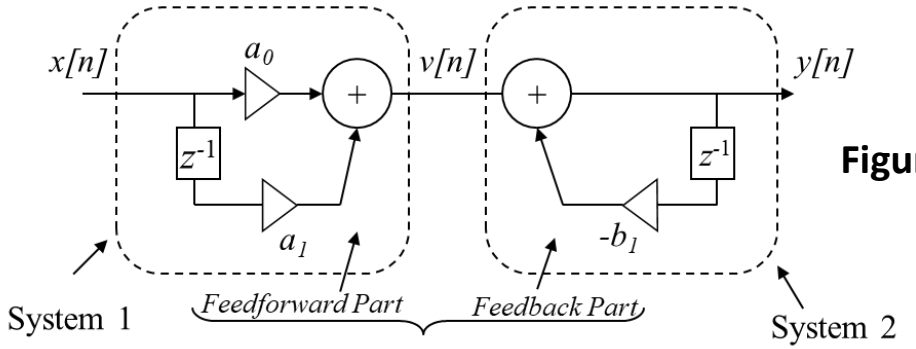


Figure 1

- Without changing the input-output relationship, we can reverse the ordering of the two systems in the cascade representation.

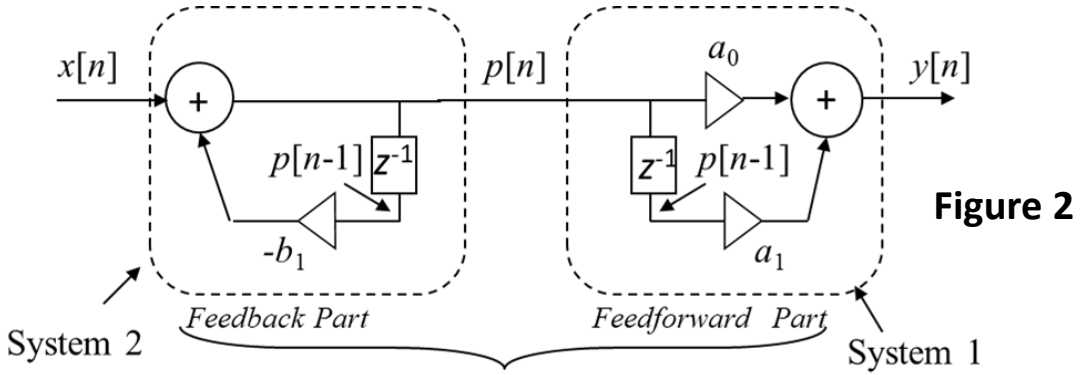
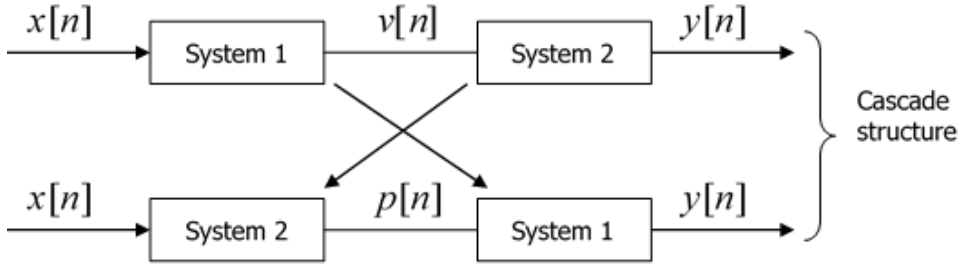


Figure 2

Note: Two systems forming a cascade structure can be interchanged without affecting the final output.



- There is no need for two delay operations; they can be combined into a single delay as shown below:

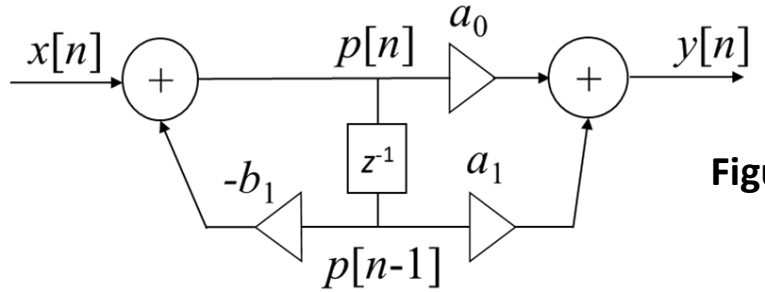


Figure 3

- It can be proven that Figures 1, 2 & 3 all represent the same difference equation:

$$y[n] = a_0 x[n] + a_1 x[n-1] - b_1 y[n-1]$$

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Classification of discrete-time systems

Classification of Systems

- In the analysis as well as in the design of systems, it is desirable to classify the systems according to the general properties they satisfy. General Categories are:
  1. Static systems
  2. Time - invariant systems
  3. Linear systems
  4. Causal systems
  5. Stable systems

1. Static systems

- A discrete-time system is called static or memoryless if its output at any instant '*n*' depends at most on the input sample at the same time, but not on past or future samples of the input.

  For example, $y[n] = ax[n]$; $y[n] = nx[n] + bx^3[n]$ are both static or memoryless
- On the other hand, the systems described by the following input-output relations are dynamic systems or system with memory:

  $y[n] = x[n] + 3x[n-1]$; $y[n] = \sum_{k=0}^{N} x[n-k]$

2. Time-invariant systems

- A time-invariant system is defined as follows:

  $x[n-n_0] \xrightarrow{H} y[n-n_0]$ where y[n] = H{x[n]}.
- Specifically, a system is time invariant if a time shift in the input signal results in an identical time shift in the output signal.

3. Linear systems

- An input $x_1[n]$ to a system (H) results in an output $y_1[n]$; Given a different input $x_2[n]$, the system output is y_2[n]. $x_1[n] \xrightarrow{H} y_1[n]$; $x_2[n] \xrightarrow{H} y_2[n]$

  For a system to be **linear**, when its input is sum $x_1(n)$ + $x_2(n)$, then the output must be the sum of the individual outputs $x_1[n] + x_2[n] \xrightarrow{H} y_1[n] + y_2[n]$
- A linear system is defined as follows:

  $a_1x_1[n] + a_2x_2[n] \xrightarrow{H} a_1y_1[n] + a_2y_2[n]$

  where a_1 and a_2 are arbitrary constants.

Classification of discrete-time systems

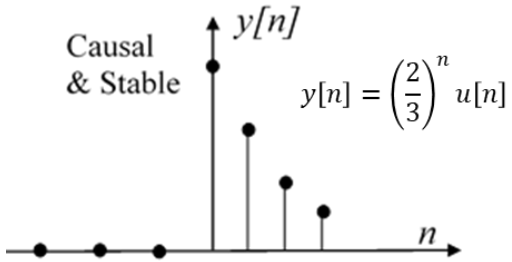
4. Causal systems

- A system is said to be causal if the output of the system at any time '*n*' depends only on present and past inputs, but does not depend on future inputs.
- If a system does not satisfy this definition, it is called *noncausal*. Such a system has an output that depends not only on present and past inputs but also on future inputs.

Examples:

$$y[n] = x[n] - x[n-1] \quad \rightarrow \text{Causal}$$
$$y[n] = ax[n] \quad \rightarrow \text{Causal}$$
$$y[n] = x[n] + 3x[n+4] \quad \rightarrow \text{Noncausal}$$
$$y[n] = x[-n] \quad \rightarrow \text{Noncausal}$$

- A discrete - time sequence is called **causal** if it has zero values for $n < 0$.



5. Stable systems

- A discrete signal *x*[*n*] is bounded if there exists a finite *M* such that $|x[n]| < M$ for all *n*.
- A discrete-time system in Bounded Input-Bounded Output (**BIBO**) stable if every bounded input sequence *x*[*n*] produced a bounded output sequence.

$$\text{If } x[n]_{max} \leq A, \text{ then } y[n]_{max} \leq B$$

Example: The following discrete-time system is at rest {i.e. *y*[-1]=0}. Check if the system is BIBO stable.

$$y[n] = ny[n-1] + x[n], \quad n > 0$$

Model answer:

If *x*[*n*]=*u*[*n*], then $|x[n]| \leq 1$. But for this bounded input, the output is unbounded (see below).

$$n = 0 \rightarrow y[0] = x[0] = 1$$
$$n = 1 \rightarrow y[1] = 1y[0] + x[1] = 2$$
$$n = 2 \rightarrow y[2] = 2y[1] + x[2] = 5$$
$$\vdots \qquad\qquad \vdots$$

y[0]=1 → y[1]=2 → y[2]=5 → increasing; Hence the system is unstable.

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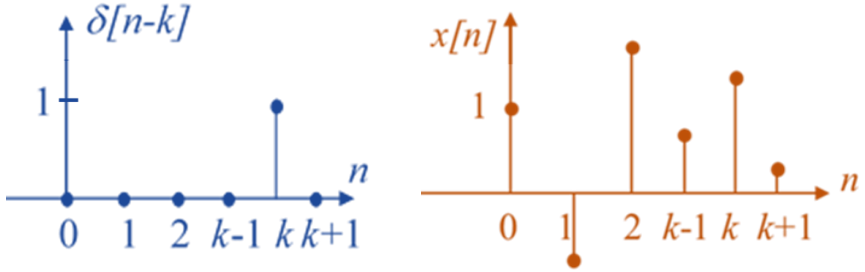
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Transformation of Discrete Time Signals

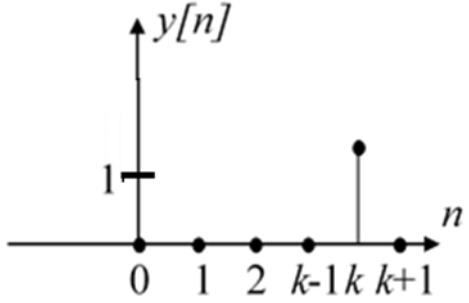
- A discrete-time signal, $x[n]$ may be shifted in time (delayed or advanced) by replacing the variables n with $n-k$ where $k > 0$ is an integer

  x[$n-k$] => $x[n]$ is delayed by k samples
  $x[n+k]$ => $x[n]$ is advanced by k samples

- For example, consider a shifted version of the unit impulse function and an arbitrary signal x[n] both shown below .

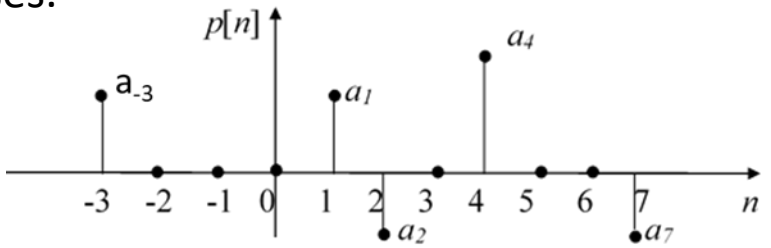


- If we multiply x[n] by $\delta[n-k]$, we obtain a signal that is zero everywhere, except at n = k.



$\therefore$ y[n] = x[n] · δ[n-k]
= x[k] · δ[n-k]

- An arbitrary sequence can then be expressed as a sum of scaled and delayed unit impulses.



$$p[n] = a_{-3}\delta[n+3] + a_1\delta[n-1] + a_2\delta[n-2] + a_4\delta[n-4] + a_7\delta[n-7]$$

- More generally, the discrete-time sequence can be expressed as:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]$$

- For a real-time signal with a finite number of samples N:

$$x[n] = \sum_{k=0}^{N-1} x[k]\,\delta[n-k]$$

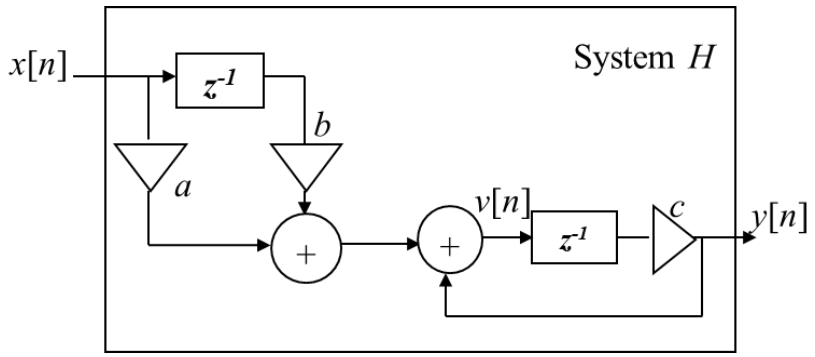
This equation is a special form of **convolution.**

- Generally, the **convolution** of two sequences $x[n]$ and $y[n]$ is defined as:

$$x[n] * y[n] = \sum_{k=-\infty}^{\infty} x[k]y[n-k] = \sum_{k=-\infty}^{\infty} x[n-k]y[k]$$

Impulse Response of a LTI System

- For example, consider the discrete-time system, H, shown below.



- The difference equation for the system *H* is given by:

$$v[n] = ax[n] + bx[n-1] + y[n]$$

$$y[n] = cv[n-1]$$

- From the above difference equations, y[n] can be determined for a given input.

Let $x[n] = \delta[n]$ (Unit impulse)

You may assume v[n] = 0 for n $\leq$ 0; therefore y[n] is also initially zero for n $\leq$ 0. Substituting n = 0, 1 ,2 ,... in the above equation , we obtain,

$n = 0$

$v[0] = ax[0]+bx[-1]+y[0] = a\cdot 1+b\cdot 0+0 = a$

$\Rightarrow \mathbf{y[1]} = c\cdot v[0] = \mathbf{a.c}$

$n = 1$

$v[1] = a\,x[1] + b\,x[0] + y[1] = a\cdot 0 + b\cdot 1 + a\cdot c = b + a\,c$

$\Rightarrow \mathbf{y[2]} = c\,v[1] = \mathbf{c\,(b+ac)}$

$n = 2$

$v[2] = a\,x[2] + b\,x[1] + y[2] = 0 + 0 + c\,(b+ac)$

$\Rightarrow \mathbf{y[3]} = c\,v[2] = \mathbf{c^2\,(b+ac)}$

.....

$n = n - 1$

$v[n\text{-}1] = c^{\,n-2}\,(b+ac) = bc^{n-2} + ac^{n-1}$

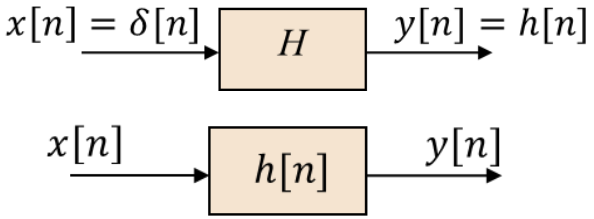
$\therefore \mathbf{y[n]} = c\,v[n\text{-}1] = \mathbf{bc^{n-1} + ac^{n}}$

$$\therefore y[n]\Big|_{x[n]=\delta[n]} = \boldsymbol{h[n]} = bc^{n-1} + ac^{n}$$

- The response $y[n] \overset{\Delta}{=} h[n]$ to an impulse excitation ($x[n] = \delta[n]$) is known as the impulse response and it is a very important characteristic of a discrete system.

Impulse Response of a LTI System

- If $x[n] = \delta[n]$, then $y[n] = h[n]$. The output tells us the system behaviour as the system, *H*, is being hit by all input frequencies. $h[n]$ completely characterizes the system, H.



- The response to an arbitrary input signal $x[n]$ is the **convolution** of $x[n]$ with the impulse response *h[n]* of the system as given by the equation below:

$$y[n] = x[n] * h[n]$$

Convolution

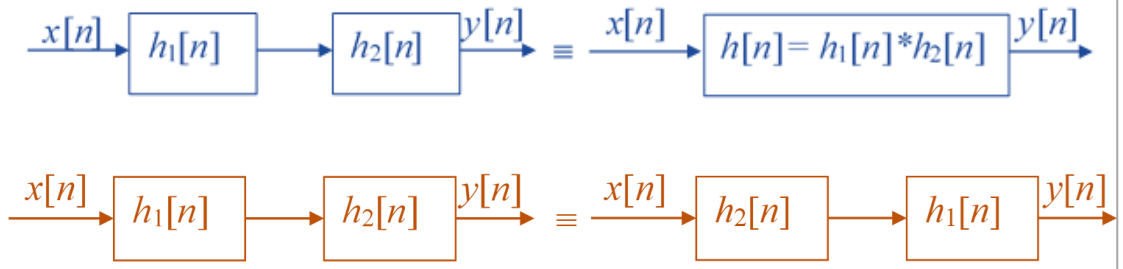
$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

$$y[n] = h[n] * x[n] = \sum_{k=-\infty}^{\infty} h[k]\, x[n-k]$$

where $h[n]=H\{\delta[n]\}$

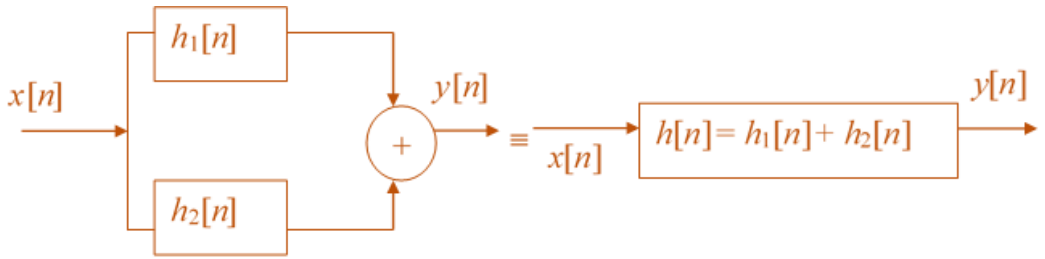
Associative Law

$$(x[n] * h_1[n]) * h_2[n] = x[n] * (h_1[n] * h_2[n])$$



Distributive Law

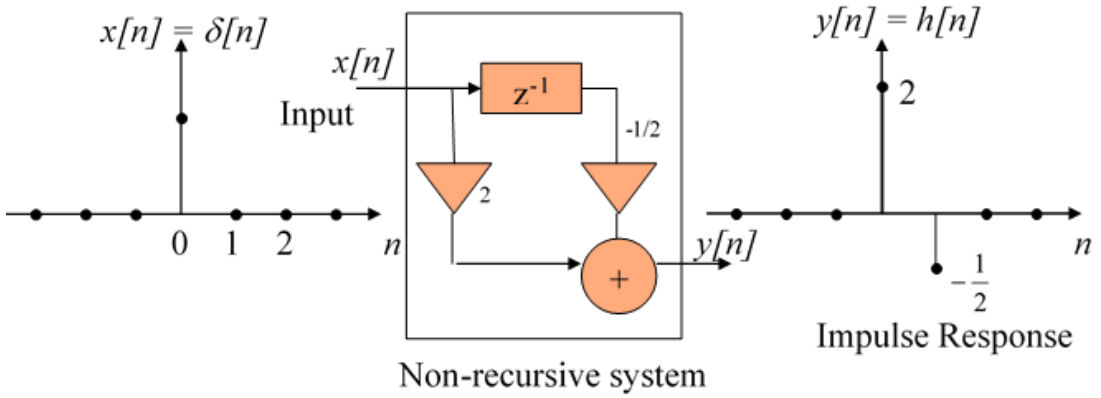
$$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$$



Impulse Response of a LTI System

Finite Impulse Response (FIR) System

- If the impulse response of a LTI system is of finite duration, the system is said to be Finite Impulse Response (FIR). It is also called **FIR filter.**



- Consider the system above; The difference equation is given by $y[n] = 2x[n] - 0.5\,x[n-1]$. Find the impulse response, $h[n]$, of the system. *Let* $x[n] = \delta[n]$

***n* = 0** $y[0] = h[0] = 2\,x[0] - 0.5\,x[-1] = 2$

***n* = 1** $y[1] = h[1] = 2\,x[1] - 0.5\,x[0] = -0.5$

***n* = 2** $y[2] = h[2] = 2x[2] - 0.5x[1] = 0$

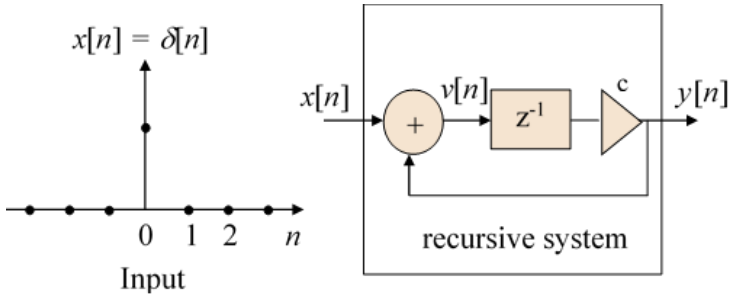
……

$n = n$ $y[n] = 0$

$$h[n] = \{2\,,\,0.5\}$$
$$h[n] = 0 \quad n \geq 2$$

Infinite Impulse Response (IIR) System

- If the impulse response of a linear time-invariant system is of infinite duration, the system is said to be an Infinite Impulse Response (IIR) system. It is also called **IIR filter.**



- Consider the system above; The difference equation is given by $v[n] = x[n] + y[n]$; $y[n] = c.v[n-1]$ ($c < 1$) Find the impulse response, $h[n]$, of the system. *Let* $x[n] = \delta[n]$

***n* = 0** $v[0] = x[0]+y[0] = 1$; ∴ $\boldsymbol{y[1]} = c \cdot v[0] = \boldsymbol{c}$

***n* = 1** $v[1] = x[1] + y[1] = c$; ∴ $\boldsymbol{y[2]} = c\,v[1] = \boldsymbol{c^2}$

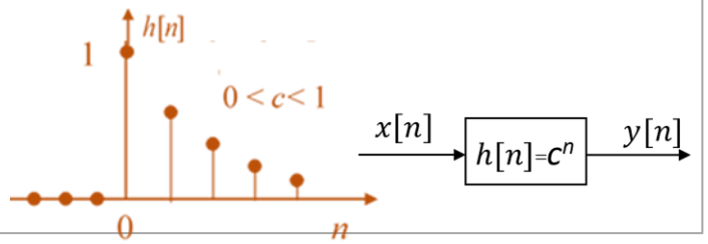
***n* = 2** $v[2] = x[2] + y[2] = c^2$; ∴ $\boldsymbol{y[3]} = c\,v[2] = \boldsymbol{c^3}$

……

***n* = *n* – 1** $v[n-1] = c^{n-1}$;

∴ $\boldsymbol{y[n]} = c\,v[n-1] = \boldsymbol{c^n}$

∴ $\boldsymbol{h[n]} = c^n$ for $n \geq 0$



Examples

Example 1:

- Find the impulse response $h[n]$ of the following fourth order non-recursive system.

$$y[n] = a_0x[n] + a_1x[n-1] + a_2x[n-2] + a_3x[n-3] + a_4x[n-4]$$

Answer: To find $h[n]$, we let $x[n] = \delta[n]$.

n=0 ➔ $h[0] = a_0\delta[0] + a_1\delta[-1] + a_2\delta[-2] + a_3\delta[-3] + a_4\delta[-4] = a_0$

n=1 ➔ $h[1] = a_0\delta[1] + a_1\delta[0] + a_2\delta[-1] + a_3\delta[-2] + a_4\delta[-3] = a_1$

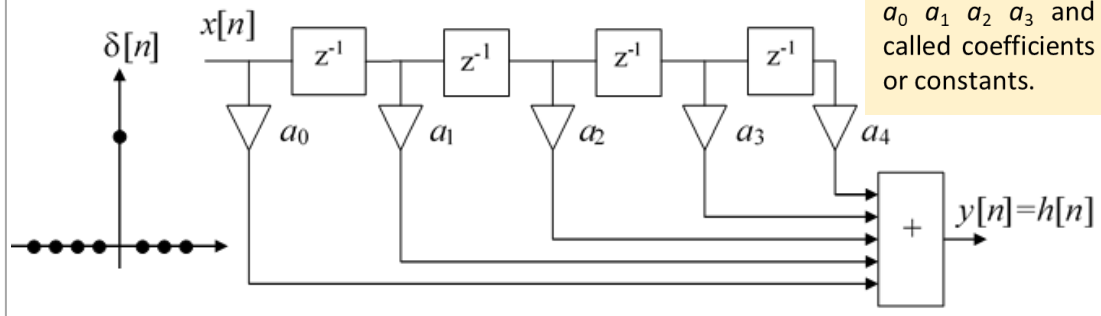
n=2 ➔ $h[2] = a_0\delta[2] + a_1\delta[1] + a_2\delta[0] + a_3\delta[-1] + a_4\delta[-2] = a_2$

n=3 ➔ $h[3] = a_0\delta[3] + a_1\delta[2] + a_2\delta[1] + a_3\delta[0] + a_4\delta[-1] = a_3$

n=4 ➔ $h[4] = 0 + 0 + 0 + 0 + a_4\delta[0] = a_4$

n=5 ➔ $h[5] = 0 + 0 + 0 + 0 + a_4\delta[1] = 0$

For n ≥ 5, h[n] = 0, since the nonzero value of δ[n] has moved out of the memory of this system.

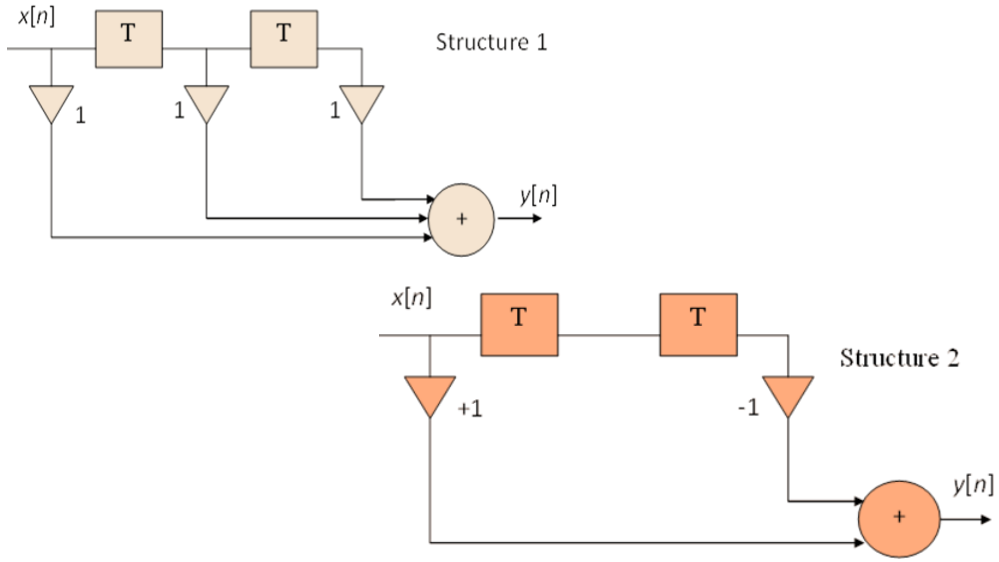


a_0 a_1 a_2 a_3 and a_4 are called coefficients (+ or -) or constants.

$h[n] = \{a_0 \;\; a_1 \;\; a_2 \;\; a_3 \;\; a_4\}$ and $h[n] = 0 \;\; n \geq 5$

Example 2:

- Two FIR systems (Structures) are shown below:
  (a) Write the difference equation;
  (b) Calculate the impulse response.



Structure 1:

$y[n] = x[n] + x[n-1] + x[n-2]$;

$h[n] = \{1\ 1\ 1\}$ and $h[n] = 0 \quad n \geq 3$

Structure 2:

$y[n] = x[n] - x[n-2]$

$h[n] = \{1\ \ 0\ \ -1\}$ and $h[n] = 0 \quad n \geq 3$

ELEC3104: Digital Signal Processing

Topic 2: Discrete-Time Systems (pp. 2-5)

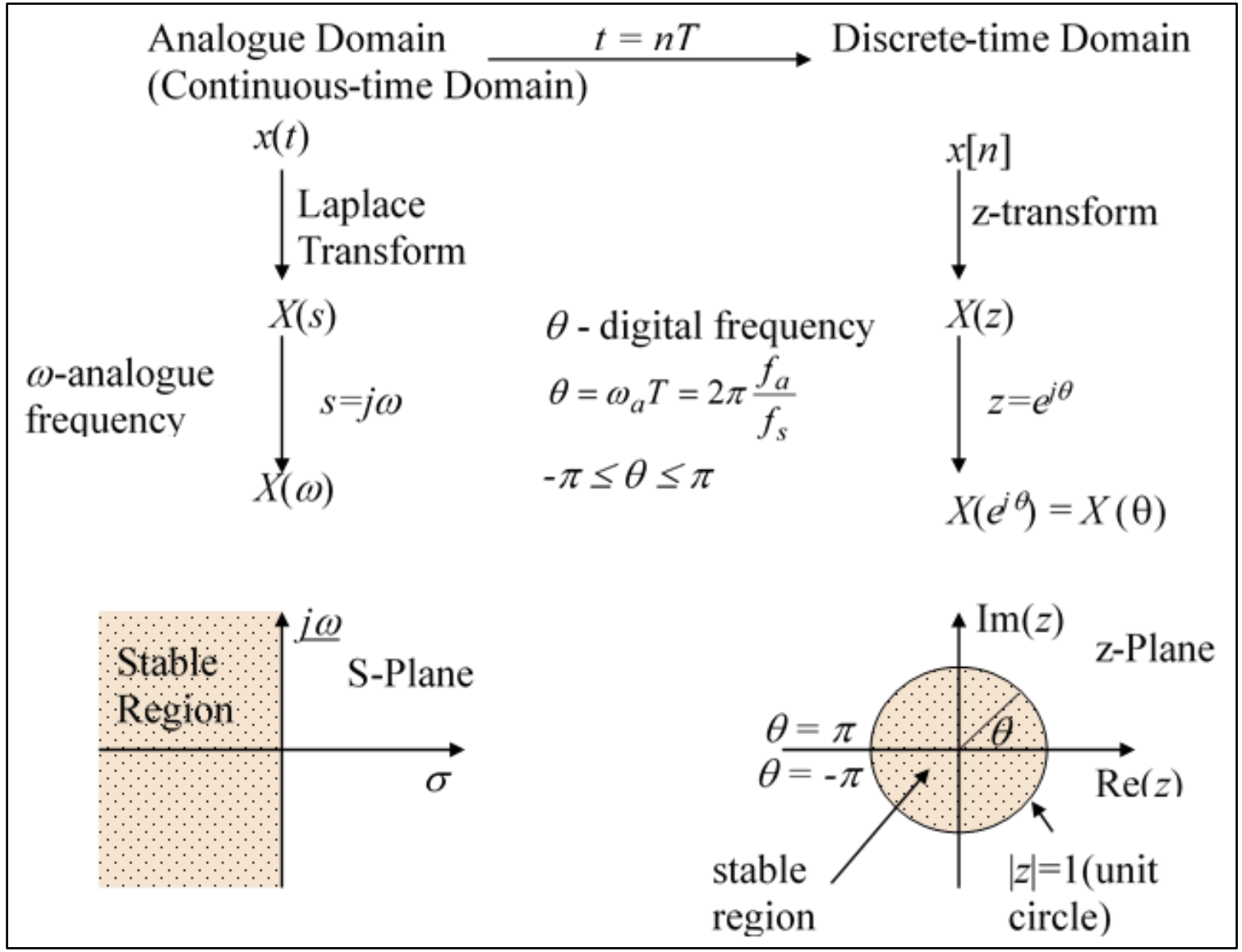
- Classification of Discrete-Time Systems (pp. 6-8)
- Linear Time Invariant (LTI) Systems (pp. 9-14)
- Introduction to z-Transform (pp. 15-23)
- Pole-Zero Description of Discrete-Time Systems (pp. 24-28)
- Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT) (pp. 29-42)
- Tutorial questions (pp. 43-47)

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Term 3, 2021

z-Transform

- The primary role of the Laplace transform in engineering is transient and stability analysis of causal LTI system described by differential equations.
- The primary roles of the z-transform are the study of system characteristics and derivation of computational structures for implementing discrete-time systems on computers.
- The z-transform is also used solve difference equations.
- We concentrate on the operational properties of the z-transform and its applications in system Analysis.



z-Transform

- The **two-sided z-transform** of a discrete-time signal is given by :

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

$z = re^{j\theta}$, θ = digital frequency; where z is a complex variable

- In causal systems, $x[n]$ may be zero when $n < 0$. The **one-sided z-transform** of a discrete-time signal is given by :

$$X(z) = \sum_{n=0}^{\infty} x[n]z^{-n}$$

- Clearly, the z-transform is a power series with an infinite number of terms and so may not converge for all values of z.

- The region where the z-transform converges is known as the **region of convergence (ROC)** and in this region the values of $X(z)$ are finite.

- **The step sequence:** $x[n] = \begin{cases} 1 & 0 \le n \le \infty \\ 0 & n < 0 \end{cases}$

$$X(z) = \sum_{n=0}^{\infty} 1 \cdot z^{-n} = 1 + z^{-1} + z^{-2} + \cdots = \frac{1}{1 - z^{-1}} = \frac{z}{z-1}$$

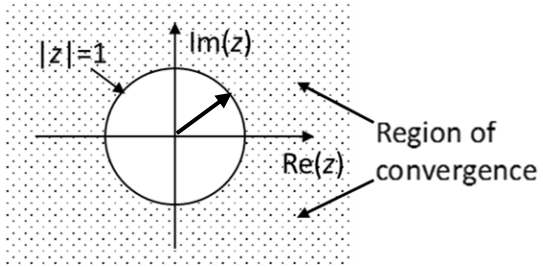
This is a geometric series with a common ratio of z^{-1}. The series converges if $|z^{-1}| < 1$ or equivalently if $|z| > 1$ and diverges if $|z| < 1$.

Let z =2 , $X(z) = 1 + 2^{-1} + 2^{-2} + \ldots = \frac{1}{1-2^{-1}} = 2$

Let $z = 0.5$, $X(z) = 1 + \left(\frac{1}{2}\right)^{-1} + \left(\frac{1}{2}\right)^{-2} + \cdots = 1 + 2 + 4 + \cdots$

- In this case, the z-transform is valid everywhere outside a circle of unit radius whose centre is at the origin (see below):

$|z|=1$ is a circle of unit radius referred to as the 'unit circle'



- The Region of Convergence (ROC) is seen to be bounded by the circle $|z|=1$, the radius of the **pole** of $X(z)\left\{= \frac{z}{z-1}\right\}$.

z-Transform

The delta sequence: δ[n]

$$Z\{\delta[n]\} = \sum_{n=0}^{\infty} \delta[n] z^{-n} = 1$$

The geometric sequence: $x[n] = a^n$

$$X(z) = \sum_{n=0}^{\infty} a^n z^{-n} = \frac{1}{1-\frac{a}{z}} = \frac{z}{z-a} \qquad \left|\frac{a}{z}\right| < 1$$

or equivalently |z| > |a|

when $a = 1$, $x[n] = 1$ for $n \geq 0$ ie. $x[n] = u[n]$

$$X(z) = \frac{z}{z-1} \qquad \text{ROC: } |z|>1$$

The complex exponential sequence: $x[n] = e^{jn\theta}$

$$Z\{e^{j\theta n}\} = \sum_{n=0}^{\infty} e^{j\theta n} z^{-n} = \frac{1}{1-\frac{e^{j\theta}}{z}} = \frac{z}{z-e^{j\theta}}$$

$$\therefore Z\{\cos n\,\theta\} = \frac{z(z-\cos\theta)}{z^2 - 2z\cos\theta + 1}$$

$$\therefore Z\{\sin n\,\theta\} = \frac{z\sin\theta}{z^2 - 2z\cos\theta + 1}$$

Properties of z-Transform

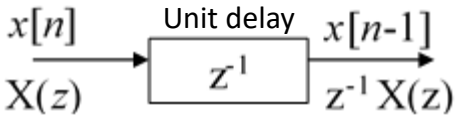
Linearity: $ax[n] + by[n] \overset{Z}{\longleftrightarrow} aX(z) + bY(z)$

Shifting property (Delay theorem):

$$x[n-k] \overset{Z}{\longleftrightarrow} z^{-k}X(z)$$

- A very important property of the z-transform is the delay theorem.
- Example: (1) $Z\{x[n-1]\} = z^{-1}X(z)$

  (2) $Z\{x[n-2]\} = z^{-2}X(z)$



Time reversal (even Symmetry): $x[-n] \overset{Z}{\longleftrightarrow} X\left(\frac{1}{z}\right) = X(z^{-1})$

Time reversal (odd Symmetry): $-x[-n] \overset{Z}{\longleftrightarrow} -X\left(\frac{1}{z}\right) = -X(z^{-1})$

Example: Find the z-transform of $x[n] = u[-n]$.

$$u[n] \overset{Z}{\longleftrightarrow} \frac{z}{z-1}; \quad u[-n] \overset{Z}{\longleftrightarrow} \frac{z^{-1}}{z^{-1}-1} = \frac{1}{1-z}$$

ROC: $|z| < 1$

z-Transform

Properties of z-Transform

Multiplication by exponential sequence:

$$a^n x[n] \overset{z}{\longleftrightarrow} X(a^{-1}z)$$

As a special case if $x[n]$ is multiplied by $e^{jn\theta}$

$$e^{j\theta n} x[n] \overset{z}{\longleftrightarrow} X(e^{-j\theta}z)$$

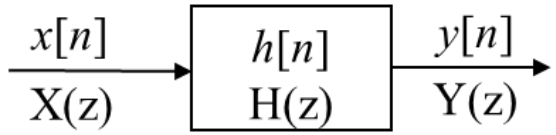
Differentiation in the z-domain:

$$nx[n] \overset{z}{\longleftrightarrow} -z\frac{d}{dz}X(z)$$

Discreate convolution:

If $y[n] = x[n] * h[n]$

then $Y(z) = X(z) \cdot H(z)$



- **Convolution** operation in time domain transforms to a product in the z-domain.

Example 3:

- **Discrete convolution in z-domain:** Compute the convolution $y[n]$ of the discrete-time signals given by (assume that both sequences start at n=0]:

$x_1[n] = [1, -2, 1]$; $\{x_1[0] = 1, x_1[1] = -2, x_1[1] = -2\}$

$x_2[n] = 1$ for $0 \le n \le 5$,

$x_2[n] = 0$ elsewhere

Ans: $y[n] = x_1[n] * x_2[n] \Rightarrow Y(z) = X_1(z) \cdot X_2(z)$

$X_1(z) = 1 - 2z^{-1} + z^{-2}$ or $X_1(z) = (1 - z^{-1})^2$

$X_2(z) = 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4} + z^{-5}$ or $X_2(z) = \frac{1-z^{-6}}{1-z^{-1}}$

$Y(z) = X_1(z) \cdot X_2(z)$

$= 1 - z^{-1} - z^{-6} + z^{-7}$

↓ Inverse z-transform

$y[n] = [1, -1, 0, 0, 0, 0, -1, 1]$

Note:

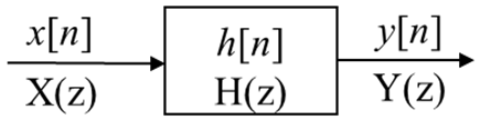
Length of $y[n]$ = Length of $x_1[n]$ + Length of $x_2[n]$ - 1

[8 samples = 3 samples + 6 samples - 1]

z-Transform

Example 4:

- Convolution of a Finite Sequence: The convolution of two finite-length sequences is also of finite length. Find the convolution of $h[n]$ = [2, 5, 0, 4] and $x[n]$ = [4, 1, 3];



If $y[n] = x[n] * h[n]$
then $Y(z) = X(z) \cdot H(z)$

Method1: Convolution using z-transform

$Y(z) = X(z) \cdot H(z)$

$X(z) = 4 + 1\,z^{-1} + 3z^{-2}$

$H(z) = 2 + 5z^{-1} + 4\,z^{-3}$

$Y(z) = 8 + 22z^{-1} + 11z^{-2} + 31z^{-3} + 4z^{-4} + 12z^{-5}$

↓ Inverse z-transform

$y[n] = [8 \quad 22 \quad 11 \quad 31 \quad 4 \quad 12]$

Method 2: Convolution in Discrete-time domain:

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k]$$

- The flipped sequence is $h[-i]$ = [4, 0, 5, 2]. Line up flipped sequence below x[n] to begin overlap and shift it successively, summing the product sequence to obtain the discrete convolution

0 shift

```
x[n]:      4  1  3
   4  0  5  2        ← h[-i]
y[0] = Sum of products = 8
```

1 shift

```
x[n]:      4  1  3
      4  0  5  2     ← h[1-i]
y[1] = (5x4)+(2x1) = 22
```

2 shifts

```
x[n]:      4  1  3
         4  0  5  2  ← h[2-i]
y[2] = 0 + 5 + 6 = 11
```

3 shifts

```
x[n]:      4  1  3
            4  0  5  2 ← h[3-i]
y[3] = 16 + 0 + 15 = 31
```

4 shifts

```
x[n]:      4  1  3
               4  0  5  2 ← h[4-i]
y[4] = 4 + 0 = 4
```

5 shifts

```
x[n]:   4  1  3
                  4  0  5  2 ← h[5-i]
y[5] = 12
```

$\therefore y[n] = x[n] * h[n] = [8 \quad 22 \quad 11 \quad 31 \quad 4 \quad 12]$

z-Transform

Transfer Function

- A transfer function is defined as the ratio of the output *Y(z)* and the input *X(z)* or as the z-transform of the system impulse response *h[n]*.
- Given the following difference equation, find the transfer function $H(z) = \frac{Y(z)}{X(z)}$.

$$y[n] = a_0x[n] + a_1x[n-1] + a_2x[n-2] + b_1y[n-1] + b_2y[n-2]$$

Take the z-transform of both sides:

$$Y(z) = a_oX(z) + a_1z^{-1}\,\mathrm{X(z)} + a_2z^{-2}\,\mathrm{X(z)} + b_1z^{-1}\,Y(z) + b_2z^{-2}\,Y(z)$$

$$Y(z) - b_1\,z^{-1}\,Y(z) - b_2\,z^{-2}\,Y(z) = a_o\,X(z) + a_1\,z^{-1}\,\mathrm{X(z)} + a_2\,z^{-2}\,\mathrm{X(z)}$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{a_0 + a_1\,z^{-1} + a_2\,z^{-2}}{1 - b_1\,z^{-1} - b_2\,z^{-2}}$$

- The transfer function is thus a ratio of polynomials in z.
- A LTI system may be described by its **transfer function**, its **impulse response**, its **difference equation** or its **poles** (roots of the denominator) and **zeros** (roots of the numerator)
- Given one form, it is possible to obtain any of the other forms.

Example 5:

- Find the difference-equation of the following transfer function: $H(z) = \frac{5z+2}{z^2+3z+2}$

Step1: First rewrite *H*(*z*) as a ratio of polynomials in z^{-1}

$$H(z) = \frac{Y(z)}{X(z)} = \frac{5z^{-1} + 2z^{-2}}{1 + 3z^{-1} + 2z^{-2}}$$

$$Y(z) + 3\,Y(z)z^{-1} + 2\,Y(z)z^{-2} = 5\,\mathrm{X(z)}\,z^{-1} + 2\mathrm{X(z)}\,z^{-2}$$

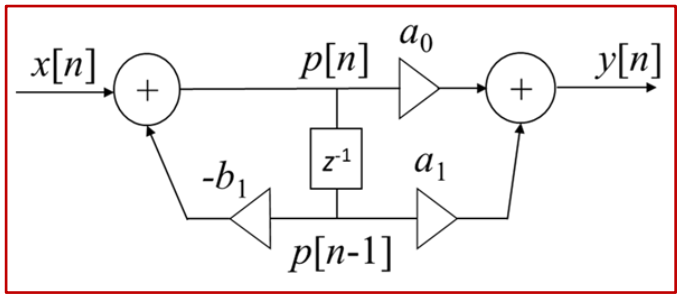
Step 2: Take inverse z-transform

$$y[n] + 3y[n-1] + 2y[n-2] = 5x[n-1] + 2x[n-2]$$

$$y[n] = 5x[n-1] + 2x[n-2] - 3y[n-1] - 2y[n-2]$$

Example 6:

Determine the Transfer Function H(z):



$$\frac{P(z)}{X(z)} = \frac{1}{1 + b_1z^{-1}}$$

$$\frac{Y(z)}{P(z)} = a_0 + a_1z^{-1}$$

$$\frac{Y(z)}{X(z)} = \frac{a_0 + a_1z^{-1}}{1 + b_1z^{-1}}$$

z-Transform

Basic z-Transform

Signal	Transform	ROC
$\delta[n]$	1	all z
$U[n]$	$\frac{1}{1-z^{-1}}$	$\|z\| > 1$
$\alpha^n u[n]$	$\frac{1}{1-\alpha z^{-1}}$	$\|z\| > \|\alpha\|$
$n\alpha^n u[n]$	$\frac{\alpha z^{-1}}{(1-\alpha z^{-1})^2}$	$\|z\| > \|\alpha\|$
$\cos(n\theta)\, u[n]$	$\frac{1-z^{-1}\cos\theta}{1-z^{-1}2\cos\theta+z^{-2}}$	$\|z\|>1$
$\sin(n\theta)\, u[n]$	$\frac{z^{-1}\sin\theta}{1-z^{-1}2\cos\theta+z^{-2}}$	$\|z\| > 1$
$r^n\cos(n\theta)\, u[n]$	$\frac{1-z^{-1}r\cos\theta}{1-z^{-1}2r\cos\theta+r^2z^{-2}}$	$\|z\| > r$
$r^n\sin(n\theta)\, u[n]$	$\frac{1-z^{-1}r\sin\theta}{1-z^{-1}2r\cos\theta+r^2z^{-2}}$	$\|z\|>r$

z-Transform Properties

Signal	Transform
$x[n]$	$X(z)$
$ax[n] + by[n]$	$aX(z) + bY(z)$
$x[n-k]$	$z^{-k}X(z)$
$a^n x[n]$	$X\left(\frac{z}{a}\right)$
$x[-n]$	$X\left(\frac{1}{z}\right)$
$x[n]*y[n]$	$X(z)\cdot Y(z)$
$nx[n]$	$-z\frac{d}{dz}X(z)$

Note:

$$S_N = 1 + z^{-1} + z^{-2} + \ldots + z^{-(N-1)} = \frac{1-z^{-N}}{1-z^{-1}}$$

If $N \to \infty$ $S_\infty = \frac{1}{1-z^{-1}}$, $|z^{-1}| < 1$

Inverse z-Transform

Partial Fraction Method

- The inverse z-transform allows us to recover the discrete-time sequence.

$$x[n] = \mathbf{Z^{-1}}[X(z)]$$

where $X(z)$ is the z-transform of $x[n]$.

Example 7:

- Find $x[n]$ for the following:

$$X(z) = \frac{z^{-1}}{1 - 0.25z^{-1} - 0.375z^{-2}}$$

Answer:

$$X(z) = \frac{z}{(z-0.75)(z+0.5)} = z\left[\frac{A}{z-0.75} + \frac{B}{z+0.5}\right] = z\left[\frac{\left(\frac{4}{5}\right)}{z-0.75} + \frac{\left(-\frac{4}{5}\right)}{z+0.5}\right]$$

$$X(z) = \left[\frac{\left(\frac{4}{5}\right)}{1 - 0.75z^{-1}} - \frac{\left(\frac{4}{5}\right)}{1 + 0.5z^{-1}}\right]$$

Inverse z-transform

$$x[n] = \frac{4}{5}\left[(0.75)^n - (-0.5)^n\right] \text{ for } n > 0.$$

Example 8:

- Find h[n] for the following system and show that the system is stable if $|a|<1$:

$$H(z) = \frac{1}{1 + az^{-1}} - \frac{az^{-1}}{1 + az^{-1}}$$

Inverse z-transform

Answer:

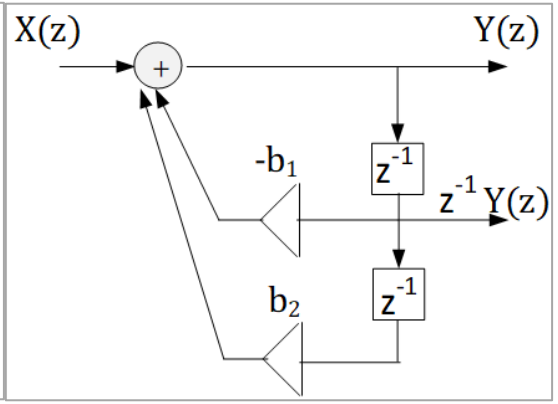
$$h[n] = \{(-a)^n u[n]\} - a\{(-a)^{n-1}\, u[n-1]\}$$

Since $|a|<1$, $(-a)^n$ and $(-a)^{n-1}$ are then decaying function. Therefore, h[n] is a decaying function. Hence the system stable.

Example 9:.

- Draw a structure for the following Transfer function:

$$H(z) = \frac{z^{-1}}{1 + b_1 z^{-1} - b_2 z^{-2}}$$



ELEC3104: Digital Signal Processing

Topic 2: Discrete-Time Systems (pp. 2-5)

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Term 3, 2021

Pole-Zero Description of Discrete Time Systems

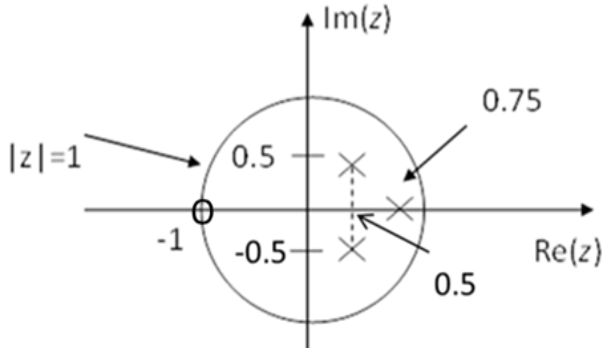
Poles and Zeros

- The **zeros** of a z-transform *H(z)* are the values of *z* for which *H(z)=0*. The **poles** of a z-transform are the values of *z* for which *H(z)=∞* . If *H(z)* is a rational function , then

$$H(z) = \frac{Y(z)}{X(z)} = \frac{a_0 + a_1 z^{-1} + \ldots + a_M z^{-M}}{1 + b_1 z^{-1} + \ldots\ldots + b_L z^{-L}}$$

$$= a_0\, z^{L-M}\, \frac{(z - z_1)(z - z_2)\ldots\ldots(z - z_M)}{(z - p_1)(z - p_2)\ldots\ldots(z - p_L)}$$

- The complex quantities (or may be real) z_1, z_2, z_3 …. are called zeros of *H(z)* and the complex quantities (or may be real) p_1, p_2, p_3 … are called the poles of *H(z)*.
- We thus see that *H(z)* is completely determined , except for the constant a_0, by the values of poles and zeros.
- The information contained in the z-transform can be conveniently displayed as a poles-zero diagram (see figure on the right)
- In the diagram, 'X' marks the position of a pole and 'O' denotes the position of a zero

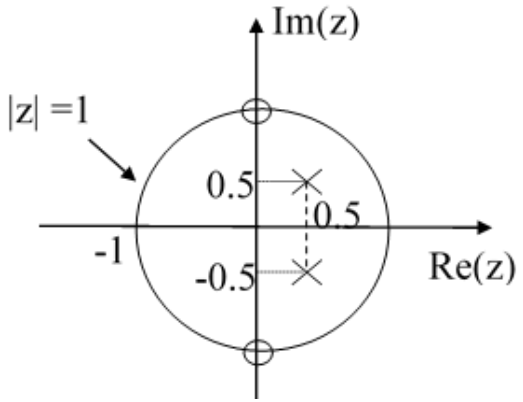


- The poles are located at $z = 0.5 \pm 0.5j$ and $z = 0.75$; a single zero is at $z = -1$.
- An important feature of the pole-zero diagram is the unit circle $|z|=1$.
- The pole-zero diagram provides an insight into the properties of a given discrete-time system.
- From the locations of the poles and zeros we can infer the frequency response of the system as well as its degree of stability.
- **For a stable system, all the poles must lie inside the unit circle. Zeros may lie inside, on, or outside the unit circle.**

Pole-Zero Description of Discrete Time Systems

Example 10:

- Determine the transfer function *H(z)* of a discrete-time system for the pole-zero diagram shown below:

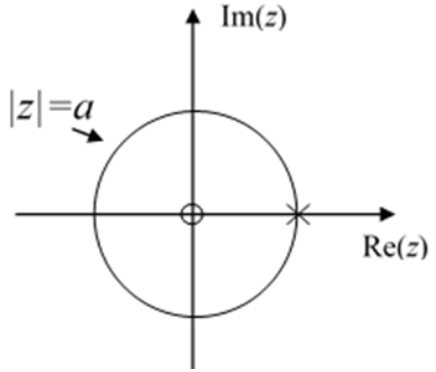


$$H(z) = \frac{K(z-j1)(z+j1)}{(z-0.5-0.5j)(z-0.5+0.5j)}$$

$$H(z) = \frac{K(1+z^{-2})}{1-z^{-1}+0.5z^{-2}}$$

Example 11:

- Determine the pole-zero plot for $H(z) = \frac{z}{z-a}$



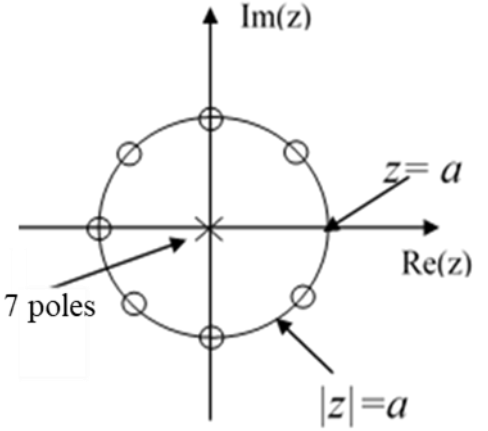
Example 12:

- Determine the pole-zero plot for

$$H(z) = \frac{z^M - a^M}{z^{M-1}(z-a)}$$

$$= \frac{(z-z_1)(z-z_2).....(z-(M-1))}{z^{M-1}}$$

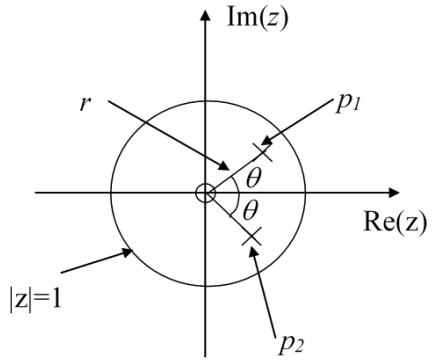
- The zero z = *a* cancels the pole at z = *a*. Thus, *H(z)* has *M*-1 zeros and *M*-1 poles as shown in the diagram below for *M* = 8.



Pole-Zero Description of Discrete Time Systems

Time domain behaviour of a system with a pair of complex conjugate poles

- Consider a system, *H(z)* with two complex conjugate poles in the z-plane:



$$p_1 = re^{j\theta}$$
$$p_2 = re^{-j\theta}$$
} Poles
$$z_1 = 0 \quad (zero)$$

- The transfer function corresponding to the pole-zero pattern:

$$H(z) = \frac{z}{(z-re^{j\theta})(z-re^{-j\theta})} = z\left[\frac{A}{z-re^{j\theta}} + \frac{B}{z-re^{-j\theta}}\right]$$

$$= z\left[\frac{\frac{1}{2jr\sin\theta}}{z-re^{j\theta}} - \frac{\frac{1}{2jr\sin\theta}}{z-re^{-j\theta}}\right] = \frac{1}{j(2r\sin\theta)}\left[\frac{1}{1-re^{j\theta}z^{-1}} - \frac{1}{1-re^{-j\theta}z^{-1}}\right]$$

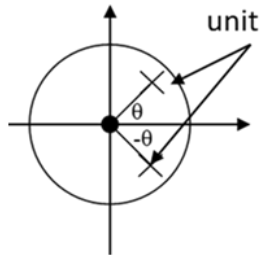
$$h(n) = \frac{1}{j2r\sin\theta}\left[(re^{j\theta})^n - (re^{-j\theta})^n\right] = \frac{r^n}{j2r\sin\theta}\left[e^{j\theta n} - e^{-j\theta n}\right]$$

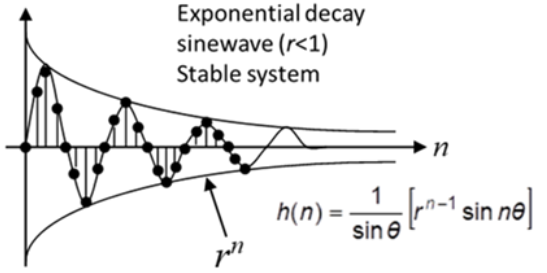
$$\boldsymbol{h(n) = \frac{1}{\sin\theta}\left[r^{n-1}\sin n\theta\right]}$$,← θ = frequency of oscillation

- This is the impulse response of the 2nd order system with complex poles. We note that the impulse response will decay away to zero provided r is less than one [$r < 1$].

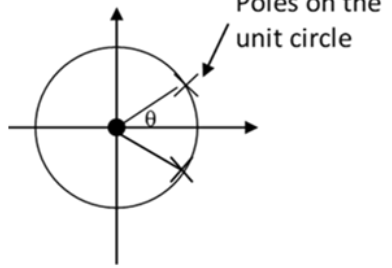
- Recall that *r* is also the distance from the origin in the z-plane to the poles p_1 or p_2, so that system will be stable if the poles in the z-plane lie inside the unit circle.

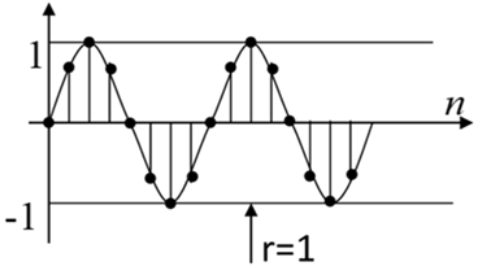
Case1:



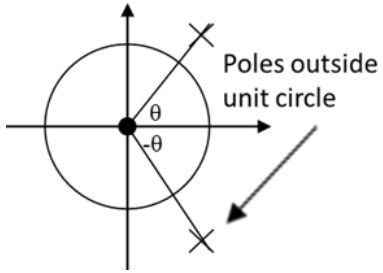


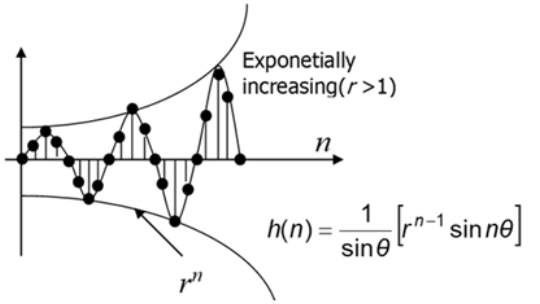
Case 2:





Case3:





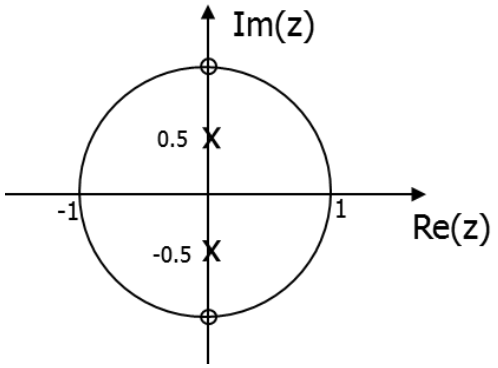
Pole-Zero Description of Discrete Time Systems

Stability Criteria:

- A system that is both **stable and causal** must have all its poles inside that unit circle within the z-plane.
- We cannot have a pole outside the unit circle, since the inverse transform of a pole located outside the circle will contribute either a right sided increasing exponential term, which **is not stable**, or a left-sided decaying exponential term that is not causal.

Example 13:

- Determine the transfer function H(z) of a discrete-time system with the pole-zero map given below:

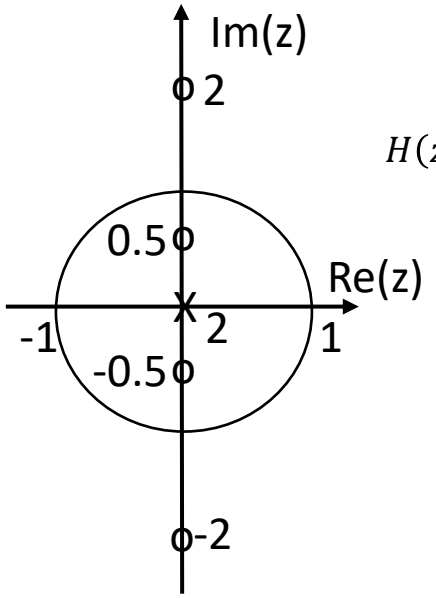


$$H(z) = K\frac{(z-j)(z+j)}{(z-0.5j)(z+0.5j)}$$

$$= K\frac{(z^2+1)}{(z^2+0.25)}$$

Example 14:

- What is the z-transform corresponding to the pole-zero pattern shown in the diagram below. Does it represent a symmetrical signal?



$$H(z) = K\frac{(z-j0.5)(z+j0.5)(z-j2)(z+j2)}{z^2}$$

$$H(z) = K\frac{(z^4+4.25z^2+1)}{z^2}$$

$$H(z) = K\,(z^2+4.25+z^{-2})$$

$$h[n] = K\,\{1, 0, 4.25, 0, 1\}$$

n = -2 n = 2

h[n] is even symmetry

- Checking for symmetry , we find that $H(z) = H(z^{-1})$ and thus h[n] s even symmetric.

ELEC3104: Digital Signal Processing

Topic 2: Discrete-Time Systems (pp. 2-5)

- Classification of Discrete-Time Systems (pp. 6-8)
- Linear Time Invariant (LTI) Systems (pp. 9-14)
- Introduction to z-Transform (pp. 15-23)
- Pole-Zero Description of Discrete-Time Systems (pp. 24-28)
- Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT) (pp. 29-42)
- Tutorial questions (pp. 43-47)

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Term 3, 2021

Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

DTFT

- The Fourier Transform is a very important tool in many aspects of engineering.
- In the case of digital signal processing, it allows signals to be represented in terms of their constituent frequency components, i.e. as signals in the 'frequency domain'.
- In some cases this means signal processing operations can be executed on frequency domain rather than time domain signals, which can have a number of advantages.
- Before proceeding, it is worth noting that in signal processing what is referred to as 'the Fourier transform' may in fact be one of four possible types of Fourier transform, illustrated in the adjacent Table.
- The difference between each type of transform is whether the time domain signal is continuous – time or discrete-time and whether its frequency domain representation is continuous (i.e. a function of real numbers ω or θ) or discrete (i.e. a function of integers n or k)

Types of Fourier transforms for signal processing

		Frequency domain signal	
		Continuous	Discrete $X[k]$
Time domain signal	Continuous $x(t)$	Fourier transform $X(\omega)$	Fourier series X_n
	Discrete $x[n]$	Discrete Time Fourier Transform (DTFT) $X(\theta)$	Discrete Fourier Transform (DFT) $X[k]$

- Fourier Transform of an arbitrary discrete signal is also called Discrete Time Fourier Transform (DTFT).
- For discrete-time signals, the Fourier transform pair is defined by:

$$X(\theta) = \sum_{n=-\infty}^{\infty} x[n]e^{-jn\theta} \quad x[n] \xleftrightarrow{\text{DTFT}} X(\theta) \quad x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\theta)e^{jn\theta}\,d\theta$$

- where θ is the digital frequency or relative frequency ($-\pi \le \theta \le \pi$) and $\theta = \omega T$. $X(\theta)$ is periodic with period 2π. (ie. $X(\theta + 2\pi) = X(\theta)$). Note that although $x[n]$ is discrete, ***X(θ)* is a continuous function** of frequency θ (hence the integral in the inverse transform).
- Note: The Fourier transform of an analogue signal is not periodic.

Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

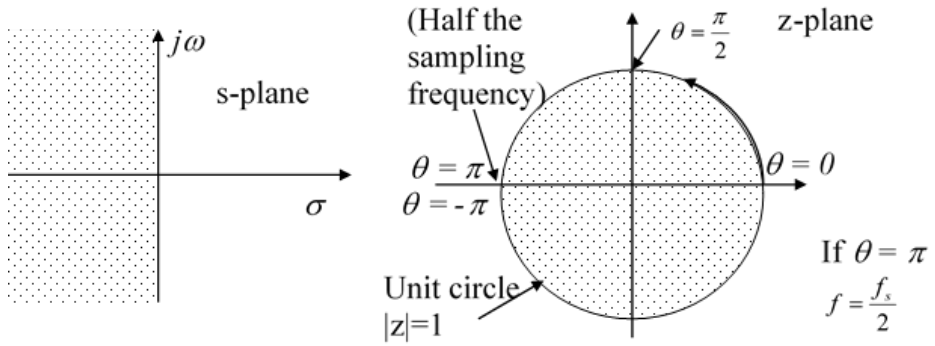
Relationship between the z-transform and Laplace transform

- If we let $z = e^{sT}$, then $z = e^{(\sigma + j\omega)T}$ (where T is the sampling period)

$$z = e^{\sigma T} \cdot e^{j\omega T} = e^{\sigma T} \cdot e^{j\theta} \qquad -\pi \le \theta \le \pi$$

$$\therefore |z| = e^{\sigma T} \text{ and } \arg(z) = \omega T = 2\pi \frac{f}{f_s} = \theta \quad (\theta = \text{digital frequency})$$

- As ω varies from -∞ to ∞ the s-plane is mapped to the z-plane as shown in figure below:



Mapping of the s-plane to the z-plane.

- The entire jω axis ($\sigma = 0$) in the s-plane is mapped onto the unit circle ($\sigma = 0$; $|z| = e^{\sigma T} = 1$)The left-hand s-plane is mapped to the inside of the unit circle and the right-hand s-plane maps to the outside of the unit circle.

- In terms of frequency response, the jω axis is the most important in the s-plane. In this case $\sigma = 0$ and the frequency points in the s-plane are related to points on the z-plane unit circle by :

$$z = e^{\sigma T} \cdot e^{j\omega T} = 1\, e^{j\theta} \qquad \boldsymbol{z = e^{j\theta}}$$

Frequency Response Estimation

- There are many instances when it is necessary to evaluate the frequency response of discrete-time systems.
- The frequency response of a system can be readily obtained from its z-transforms.
- For example, if we set $z = e^{j\theta}$, that is evaluate the z-transform around the unit circle, we obtain the Discrete Time Fourier Transform (DTFT) of the system.

DTFT: $H(\theta) = H(z)|_{z=e^{j\theta}}$ $-\pi \le \theta \le \pi$

Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

Example 15:

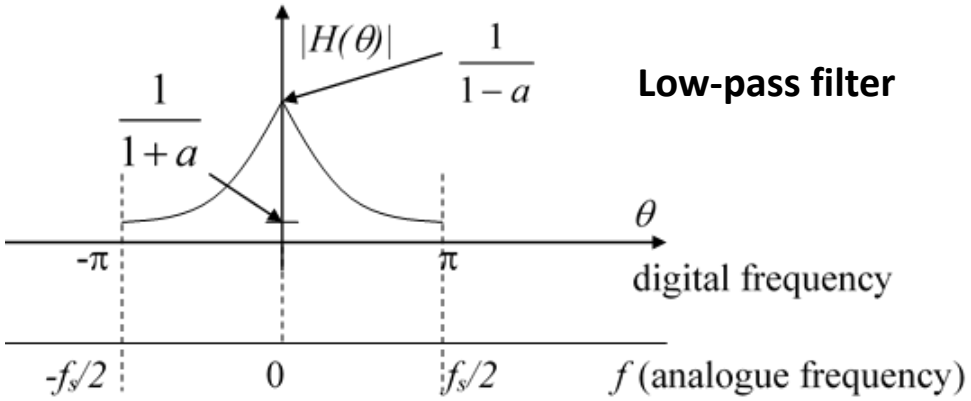
- Determine and sketch an approximate magnitude response for the following system:

$$H(z) = \frac{1}{1-az^{-1}} \qquad 0 < a < 1$$

- To find $H(\theta)$ (the frequency response) , substitute $z = e^{j\theta}$

$$H(\theta) = H(z)|_{z=e^{j\theta}} = \frac{1}{1-ae^{-j\theta}} = \frac{1}{(1-a\cos\theta)+ja\sin\theta}$$

$$|H(\theta)| = \frac{1}{\sqrt{(1-a\cos\theta)^2+(a\sin\theta)^2}} = \frac{1}{\sqrt{1-2a\cos\theta+a^2}}$$



Low-pass filter

- **DC gain or DC value : $|H(0)| = {}^{1}/_{1-a}$** {when θ=0}

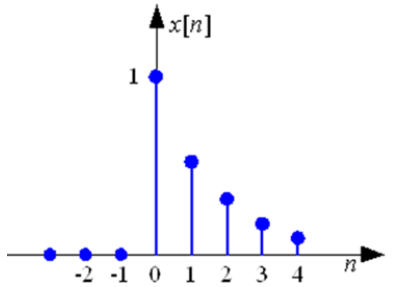
Example 16:

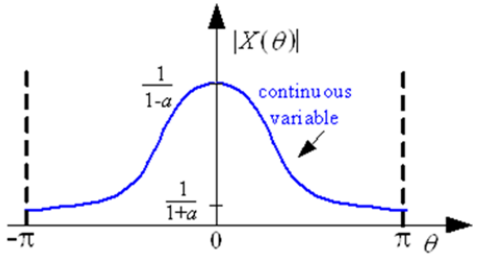
- Determine DTFT of the following sequence and sketch an approximate magnitude response.

$$x[n] = \begin{cases} 0 & n < 0 \\ a^n & n \geq 0, |a| < 1 \end{cases} \qquad X(\theta) = \sum_{n=-\infty}^{\infty} x[n]e^{-jn\theta}$$

$$X(\theta) = \sum_{n=0}^{\infty} a^n e^{-jn\theta} = \sum_{n=0}^{\infty} (ae^{-j\theta})^n = \frac{1}{1-ae^{-j\theta}}$$

$$X(\theta) = \frac{1}{(1-a\cos\theta)+ja(\sin\theta)} \quad \therefore |X(\theta)| = \frac{1}{\sqrt{1-2a\cos\theta+a^2}}$$





- *Note:* The above $X(\theta)$ can be obtained using the z-transform as well: $x[n] = a^n u[n] \Rightarrow X(z) = \frac{1}{1-az^{-1}}$

$$X(\theta) = X(z)\Big|_{z=e^{j\theta}} = \frac{1}{1-ae^{-j\theta}}$$

Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

Example 17:

- Consider a system described by $y[n] = x[n] - x[n-1]$. Find the (a) impulse response; (b) transfer function $H(z) = \frac{Y(z)}{X(z)}$; (c) Discrete time Fourier Transform $H(\theta)$. Sketch the approximate magnitude response.

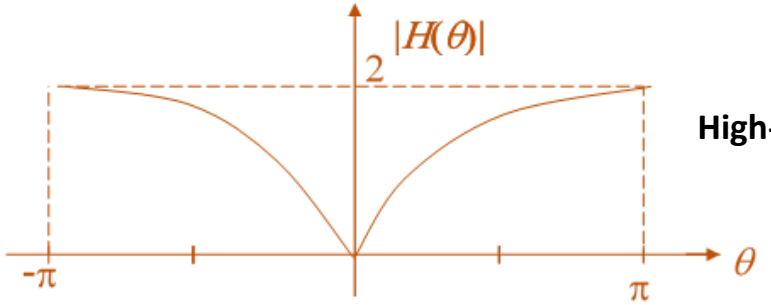
Answer:

$(a)\ h[n] = \{1, -1\}$;

$(b)\ Y(z) = X(z) - z^{-1}X(z);\ \ H(z) = 1 - z^{-1}$

$(c)\ H(\theta) = 1 - e^{-j\theta}$

$$|H(\theta)| = \sqrt{(1-\cos\theta)^2 + (-\sin\theta)^2} = \sqrt{2 - 2\cos\theta}$$



High-pass filter

- **DC gain : $|H(0)| = 0$** {when $\theta = 0$}

Properties of DTFT

$x[n] \overset{\text{DTFT}}{\rightarrow} X(\theta)$ $\quad x_1[n] \overset{\text{DTFT}}{\rightarrow} X_1(\theta)$ $\quad x_2[n] \overset{\text{DTFT}}{\rightarrow} X_2(\theta)$

- Linearity: $ax_1[n] + bx_2[n] \overset{\text{DTFT}}{\leftrightarrow} aX_1(\theta) + bX_2(\theta)$
- Time shift: $x[n-k] \overset{\text{DTFT}}{\leftrightarrow} e^{-j\theta k}X(\theta)$
- Frequency Shift: $x[n]e^{jn\theta_0} \overset{\text{DTFT}}{\leftrightarrow} X(\theta - \theta_0)$
- Frequency Differentiation: $nx[n] \overset{DTFT}{\leftrightarrow} j\frac{d}{d\theta}X(\theta)$
- Convolution: $x_1[n] * x_2[n] \overset{DTFT}{\leftrightarrow} X_1(\theta)X_2(\theta)$
- Multiplication: $x_1[n]x_2[n] \overset{DTFT}{\leftrightarrow} X_1(\theta) * X_2(\theta)$
- Parseval's Theorem:

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} |X(\theta)|^2 d\theta$$

Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

Example 18:

- The frequency response of an ideal low pass filter, in the fundamental interval $-\pi \le \theta \le \pi$ is given by

$$H(\theta) = \begin{cases} \frac{1}{2} & 0 \le |\theta| \le \theta_C \\ 0 & \theta_C < |\theta| \le \pi \end{cases}$$

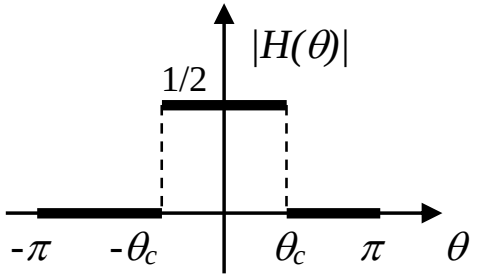
$$X(\theta) = \sum_{n=-\infty}^{\infty} x[n]e^{-jn\theta} \quad x[n] \xleftrightarrow{\text{DTFT}} X(\theta) \quad x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\theta)e^{jn\theta}\, d\theta$$

(a) Find the impulse response of the ideal low pass filter;

(b) Sketch the impulse response for $\theta_C = \frac{\pi}{4}$.

Answer:

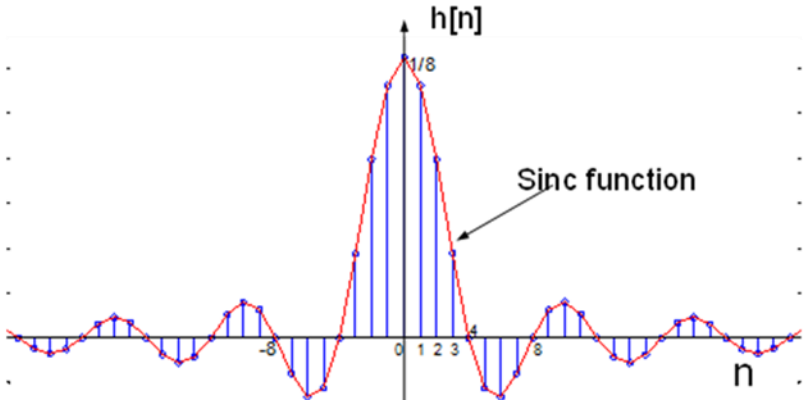
(a)



$$h[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H(\theta)e^{jn\theta}d\theta = \frac{1}{2\pi}\int_{-\theta_C}^{\theta_C}\frac{1}{2}e^{jn\theta}d\theta$$

$$= \frac{1}{4\pi}\frac{e^{j\theta_C n} - e^{-j\theta_C n}}{jn} = \frac{1}{2\pi}\frac{\sin\theta_c n}{n}$$

(b) $\theta_c = \frac{\pi}{4}$ $\quad h(n) = \frac{1}{2\pi}\frac{\sin\frac{n\pi}{4}}{n} = \frac{1}{8}\frac{\sin\frac{n\pi}{4}}{\frac{n\pi}{4}}$



Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

Discrete Fourier Transform (DFT)

- The importance of the discrete Fourier transform (DFT) for practical applications is hard to overstate.
- If you've ever seen a real time spectrum of just about anything, chances are it was generated using a DFT.
- The DFT allows us to determine the frequency content of a signal, that is, to perform spectral analysis.
- The DFT plays a central role in the implementation of a variety of digital signal processing algorithms, as a result of the existence of the efficient algorithm for the Fast Fourier Transform (FFT).

$$x[n] \xleftrightarrow{DTFT} X(\theta) \quad \text{DTFT}$$

$$x[n] \xleftrightarrow{DFT} X[k] \quad \text{DFT}$$

{DFT mapping $x[n]$ to another sequence}

- The DFT denoted by $X[k]$ (complex valued sequence), is obtained by sampling the Discrete Time Fourier Transform $X(\theta)$ at a finite number of frequency points
- This sampling is conventionally performed at equally spaced points over the period extending over $-\pi \leq \theta \leq \pi$.

(1) Fourier transform of a discrete signal (DTFT) is

$$X(\theta) = \sum_{n=-\infty}^{\infty} x[n]e^{-jn\theta}, \quad -\pi \leq \theta \leq \pi$$

(2) $X(\theta) = X(z)|_{z=e^{j\theta}}$ {Evaluate z-transform on the unit circle}

(3) Discrete Fourier Transform (DFT) or N-point DFT

DFT: $X[k] = \sum_{n=0}^{N-1} x[n]e^{-jn\left(\frac{2\pi k}{N}\right)}$

IDFT: $x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]e^{jn\left(\frac{2\pi k}{N}\right)}$

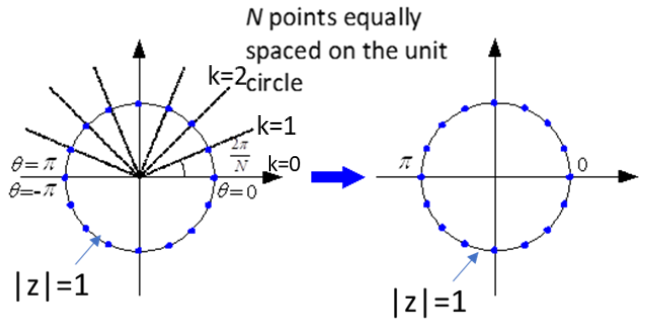
- i.e. sampling $X(\theta)$ at N equally spaced interval, k = 0, 1, 2, … N-1. Note that one of the most important properties of the DFT is implied periodicity.

Number of time samples = N
Number of frequency samples (k) = N

Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

Sampling the unit circle at N points equaly spaced in angle:

- The DFT corresponds to sampling the z-transform of *X(z)* at *N*-points equally spaced in angle around the unit circle ($\theta = \frac{2\pi k}{N}$; $(k = 0, 1, 2, \dots N-1)$

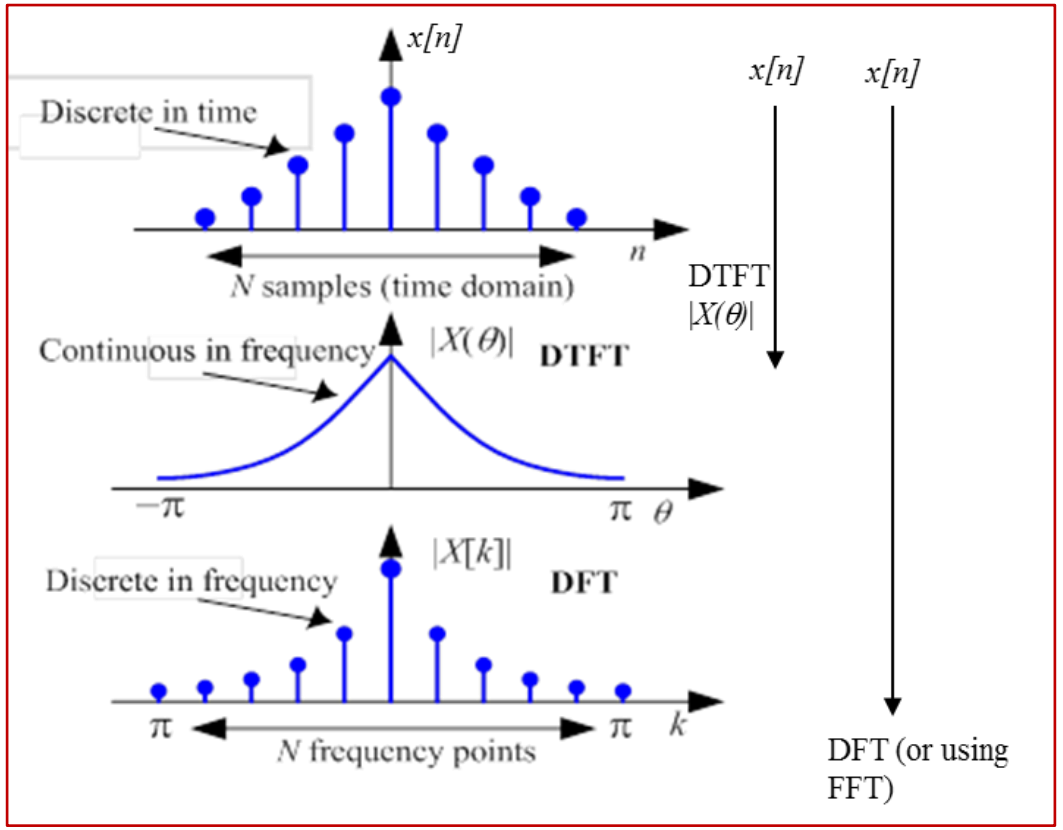


- The diagram on the right shows an example of a sequence $x[n]$ and its corresponding DTFT, X(θ), and the sampled version of DTFT known as DFT, X[k].

Approximating the DTFT by the DFT

- When the DFT of a block of *N* samples (i.e. a window of samples) is calculated, the assumption is that the original signal actually repeats itself periodically, with period *N*.
- Clearly, for all real signals this will not be true.
- Even for artificial signals such as a pure sinusoid this will only be true if *N* is a multiple of the period of the sinusoid.
- This "windowing" process introduces a slight distortion into the frequency representation of the signal being analysed.

If x[n] is a finite N-point signal with $n = 0, 1, 2 \dots . N-1$, the DFT is an exact match of its DTFT $X(\theta)$ at $\theta = \frac{2\pi k}{N}$ $(k = 0, 1, 2, \dots N-1)$ and the IDFT results in perfect recovery of x[n].



Number of time samples = *N*
Number of frequency samples (*k*) = *N*

Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

DFT calculation:

$$X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-jn\frac{2\pi k}{N}} \qquad k = 0,1,2,3,\ldots N$$

$$X[k] = \sum_{n=0}^{N-1} x[n]\cos\left(\frac{2\pi}{N}nk\right) - j\sum_{n=0}^{N-1} x[n]\sin\left(\frac{2\pi}{N}nk\right)$$

$$X[k] = A[k] - jB[k]$$

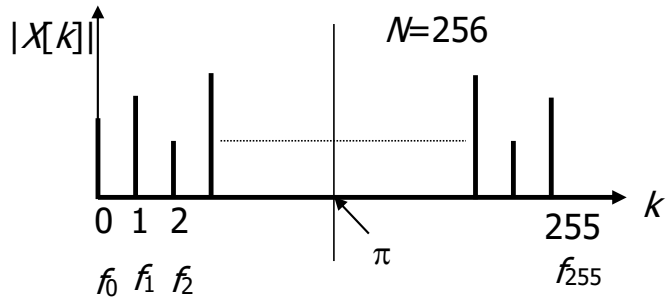
$$|X[k]| = \sqrt{A[k]^2 + B[k]^2} \qquad k = 0,1,2,3,\ldots N-1$$

$$\Delta\theta = \frac{2\pi}{N}$$

$$\Delta f = \frac{f_S}{N}$$

We have $f_k = k\left(\frac{f_s}{N}\right) = k\Delta f \quad (k = 0, 1, 2, \ldots, N-1)$

- N is the length of the sequence x[n] to be analysed using DFT and $|X[k]|$ is the magnitude of the sampled spectrum at index k.



The frequency resolution ($\Delta f = f_s/N$) can be made as small as desired by increasing the value of N.

Example 19:

- Let f_s = 8000 Hz, number of sample N = 1000
- Frequency resolution $\Delta f = \frac{f_S}{N} = \frac{8000}{1000} = 8Hz$
- $f_0 = 0$; $f_1 = 8$Hz, $f_2 = 16$ Hz, …. $f_{999} = 7992$Hz

$$X[k] = \sum_0^{999} x[n]e^{-jn\left(\frac{2\pi k}{1000}\right)}, \; k = 0, 1, 2, 3, 4..$$

Example 20:

- A speech signal is sampled at a rate of 20000 samples/sec. A sequence of length (N) 1024 samples is selected and the 1024-point DFT is computed.

(a) What is the time duration of segment of speech?

Duration = no of samples × sampling period.

= 1024 (1/20000) = 51.2 ms

(b) What is the frequency resolution (spacing in Hz) between the DFT values?

$$\text{Resolution} = \frac{f_S}{N} = \frac{20000}{1024} = 19.531Hz$$

Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

Example 21:

- Consider the finite length sequence: $x[n] = \delta[n] + 0.2\delta[n-2]$

  Find the *N*-point DFT of *x*[*n*] for *N* = 50.

Answer:

$$X(z) = 1 + 0.2\, z^{-2} \Rightarrow X(\theta) = 1 + 0.2\, e^{-j2\theta}$$

The *N*-point DFT is obtained by evaluating $X(\theta)$ at *N* points equally spaced around the unit circle: $\theta = \frac{2\pi k}{N}$

Therefore, the DFT is given by:

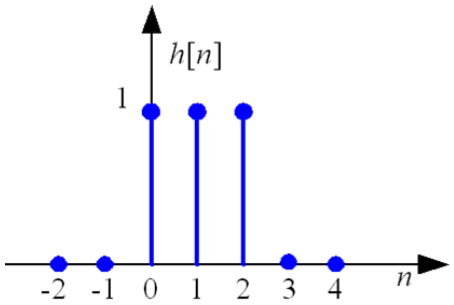
$$X[k] = 1 + 0.2e^{-j\left(\frac{2\pi k}{N}\right)2}$$

$$= 1 + 0.2e^{-j\left(\frac{2\pi k}{25}\right)} \qquad k = 0, 1, 2, \ldots. 49$$

Example 22:

- Find the DFT of the three-sample averager.

$$h[n] = \begin{cases} \frac{1}{3} & 0 \le n \le 2 \\ 0 & \text{otherwise} \end{cases}$$



$$H[k] = \sum_{n=0}^{N-1} h[n]e^{-jn(\frac{2\pi}{N}k)} = \sum_{n=0}^{2} \frac{1}{3} e^{-jn(\frac{2\pi}{N}k)}$$

$$H[k] = e^{-j(\frac{2\pi}{N}k)} \frac{(1 + 2\,cos(\frac{2\pi k}{N}))}{3}$$

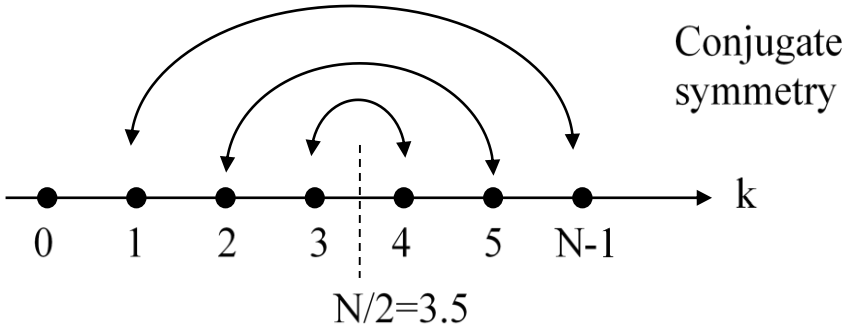
$$|H[k]| = \frac{1}{3}\left|[1 + 2\,cos(\frac{2\pi k}{N})]\right| \quad \text{where } k = 0,1,2$$

When k= 0, we obtain the DC gain or DC value. $|H[0]| = 1$

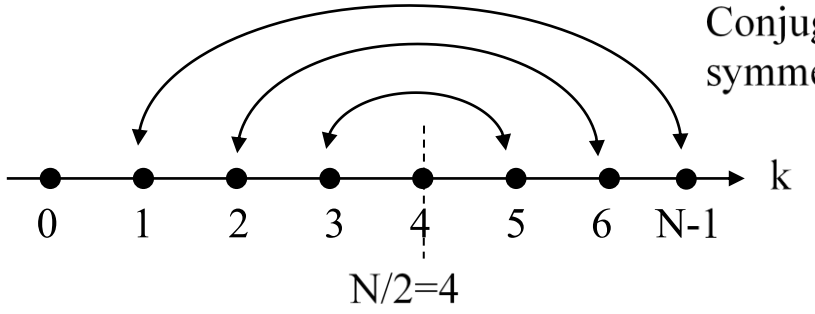
Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

Conjugate Symmetry:

- The DFT of a real sequence possesses conjugate symmetry about the origin with $X[-k] = X^*[k]$
- Since the DFT is periodic, $X[-k] = X[N-k]$
- **N is odd** (N=7)



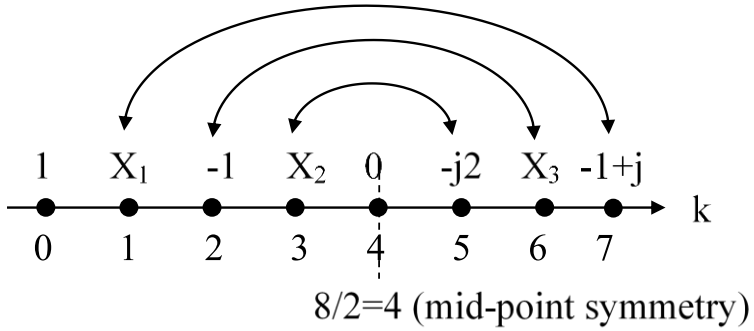
- **N is even** (N=8)



Example 23:

- The DFT of a real signal is $\{1, X_1, -1, X_2, 0, -j2, X_3, -1+j\}$

  Find X_1, X_2 and X_3.



$$X_1 = X^*[7] = -1 - j$$

$$X_2 = X^*[5] = j2$$

$$X_3 = X^*[2] = -1$$

- **Note:** 8 frequency points will result in 8 time samples when IDFT is applied.

Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

Parseval's Theorem

- DFT is an energy-conserving transformation that allows to find the signal energy from the signal or its spectrum.
- Therefore, the sum of squares of the signal samples is related to the sum of squares of the magnitude of the DFT samples:

$$\sum_{n=0}^{N-1}|x[n]|^2=\frac{1}{N}\sum_{k=0}^{N-1}|X[k]|^2$$

Example 24:

- The DFT of a real signal is: $\{1, X_1, -1, X_2, -7, -j2, X_3, -1+j\}$. What is the signal energy? (Note: First value is at k=0 and the last value is at k = 7)

Ans: X_1, X_2 and X_3 can be found first using conjugate symmetry property

$$\text{Signal Energy}=\frac{1}{8}\sum|X[k]|^2$$

$$=\frac{1}{8}\left[1^2+(\sqrt{2})^2+(-1)^2+2^2+7^2+2^2+1^2+\left(\sqrt{2}\right)^2\right]$$

$$\therefore \text{Signal Energy}=8$$

Inverse Discrete Fourier Transform (IDFT)

- The inverse DFT equation comes directly from the Inverse Fourier Transform equation

$$x[n]=\frac{1}{N}\sum_{k=0}^{N-1}X(k)e^{jn\frac{2\pi}{N}k}\quad 0\le n\le N-1$$

However, the signal which it produces will be a periodic signal repeating every *N* samples

Example 25:

- If $X[k]$ = {4, -*j*2, 0, *j*2}, find its IDFT.

Answer:

N=4, $\therefore x[n]=\frac{1}{4}\sum_{k=0}^{3}X[k]e^{j\frac{2\pi kn}{4}}=\frac{1}{4}\sum_{k=0}^{3}X[k]e^{j\frac{\pi kn}{2}}$

- $n=0\quad x[0]=\frac{1}{4}[4-j2+0+j2]=1$
- $n=1\quad x[1]=\frac{1}{4}[4-j2e^{j\frac{\pi}{2}}+0+j2e^{j\frac{3\pi}{2}}]=2$
- $n=2\quad x[2]=\frac{1}{4}\left[4-j2e^{j\pi}+0+j2e^{j3\pi}\right]=1$
- $n=3\quad x[3]=\frac{1}{4}[4-j2e^{j\frac{3\pi}{2}}+0+j2e^{j\frac{9\pi}{2}}]=0$
- The IDFT is $x[n]$={1, 2, 1, 0}(Note: When $n=0, x[0]=1$)

Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

Padding with Zeros and Frequency Resolution

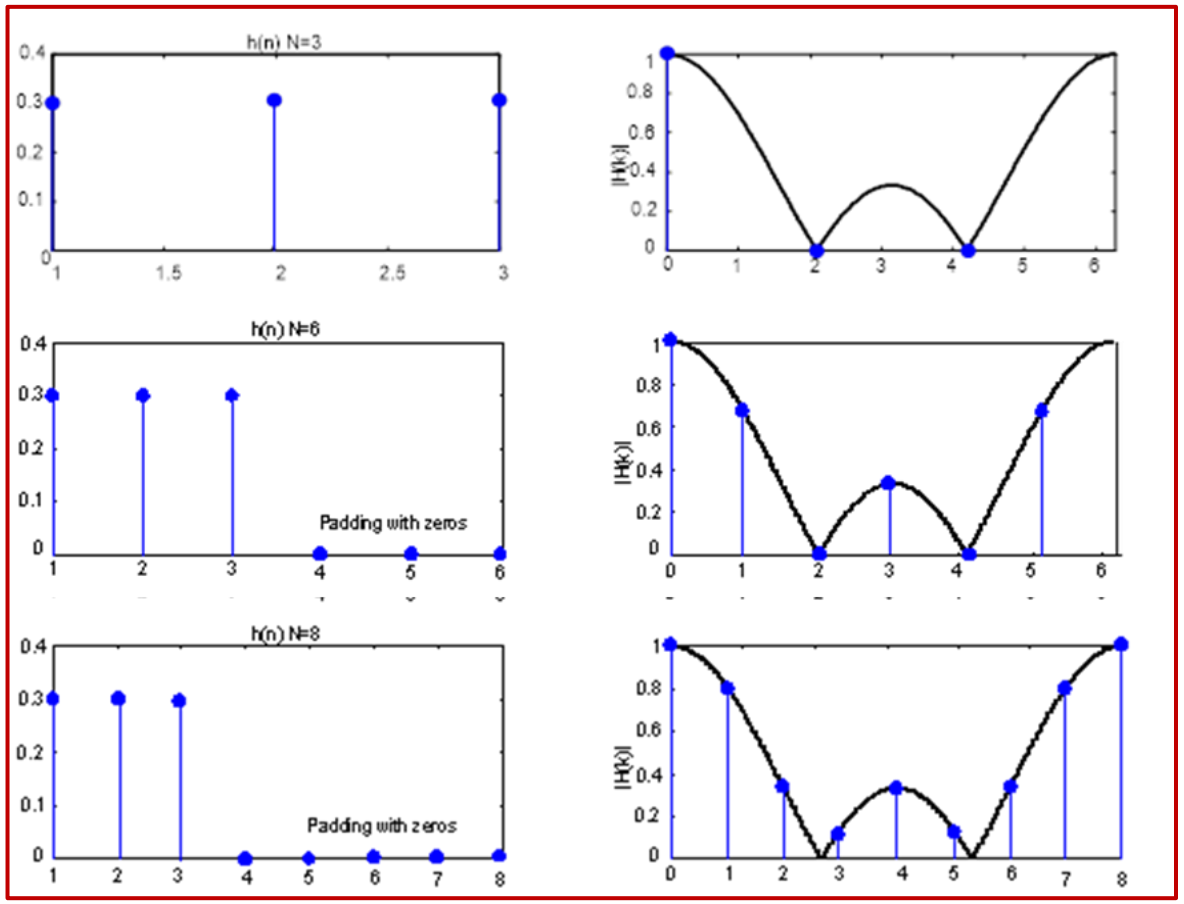
- DFT: $X[k] = \sum_{n=0}^{N-1} x[n] e^{-jn\left(\frac{2\pi k}{N}\right)}$
- The frequency resolution depends directly on the number of time samples *N*, i.e. the longer the sequence *x*[*n*], the more frequency points in the DFT domain (frequency domain).
- To obtain more points in the DFT sequence, we can always increase the duration of *x[n]* by adding additional zero-valued elements. This procedure is called padding with zeros.
- These zero-valued elements contribute nothing to the sum in the above equation, but act to decrease the frequency spacing ($2\pi/N$).
- The zero padding gives us a high-density spectrum and provided a better displayed version for plotting. But it does not give us a high resolution spectrum because no new information is added to the signal. Only additional zeros are added in the data.

Example 26:

- The three sample averager can be transformed using both the DTFT and the DFT. Because the DTFT is determined for all $n \in [-\infty,\infty]$, it is unaffected by the window length *N*.
- On the other hand, adding zeros to the three sample average (Example 22) gives a different view of the magnitude spectrum, depending on the length *N*.
- Lengths of *N* = 3, 6 and 8 are shown on the next page .
- Note that although the frequencies at which the DFT is evaluated are different in each case, the underlying spectrum (given by the DTFT) is exactly the same.

Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

Three Sample Averager



Decreasing spectral spacing by zero padding to increase the number of samples. Padding with 3 zeros (N=6) and padding with 5 zeros (N=8)

Fast Fourier Transform (FFT)

- The Fast Fourier Transform (FFT) is simply a mathematical technique to accelerate the calculation of the DFT. It was developed by Cooley and Tukey (1965) requires *N* to be a power of 2.
- Typically, if the DFT is calculated for a block of 2*n* samples e.g. 512 or 1024 samples (*N*) it would make the calculation of the DFT quite demanding.
- The FFT simply uses repetition and redundancy in the calculation to speed it up.
- The FFT is simply a TECHNIQUE to calculate the DFT, NOT a different transform.

ELEC3104: Digital Signal Processing

Topic 2: Discrete-Time Systems (pp. 2-5)

- Classification of Discrete-Time Systems (pp. 6-8)
- Linear Time Invariant (LTI) Systems (pp. 9-14)
- Introduction to z-Transform (pp. 15-23)
- Pole-Zero Description of Discrete-Time Systems (pp. 24-28)
- Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT) (pp. 29-42)
- Tutorial questions (pp. 43-47)

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Term 3, 2021

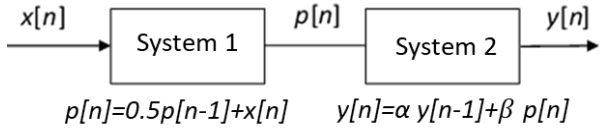
Topic 1: Tutorial Questions (Level 2)

SOLO – Level 2

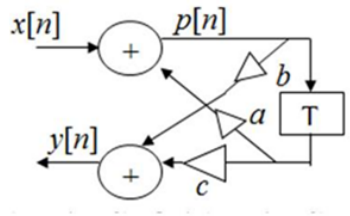
1. Consider the cascade of the following two systems as depicted below. The difference equation relating $x[n]$ of $y[n]$ is given by:

$$y[n] = x[n] + \frac{3}{4}y[n-1] - \frac{1}{8}y[n-2]$$

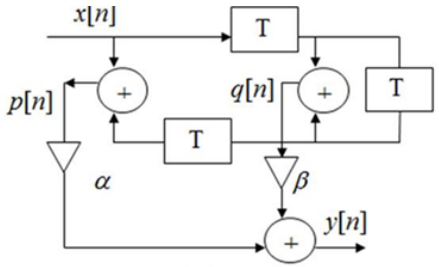
Determine α and β.



2. For the block diagram realisations given below, develop the relation between y[n] and x[n] in each case.



(a)



(b)

3. Draw a system implementation for each of the following equations:

a) $2y[n] + y[n-1] - 4y[n-3] = x[n] + 3x[n-5]$

b) $y[n] = x[n] - x[n-N]$

c) $y[n] = a_0x[n] + a_1x[n-1] + a_2x[n-2] - b_1y[n-1] - b_2y[n-2]$

4. A difference equation for a particular filter is given by

$$y[n] = 0.1x[n] - 0.1x[n-2] + 0.8x[n-3] + 0.1x[n-4] + 0.6x[n-6]$$

Find the impulse response of the above filter.

SOLO – Level 2

5. Consider the system described by the difference equation

$$y[n] = ay[n-1] + bx[n]$$

Determine '*b*' in terms of '*a*' so that $\sum_{n=-\infty}^{\infty} h[n] = 1$.

6. Let $x[n]$ = {1 4 0 2} and $h[n]$ = {1 2 1}. Find their convolution (Both sequences start at n=0).

7. Find the impulse response h[n] for the following system:

$$H(z) = \frac{1 - \frac{1}{2}z^{-1}}{\left[1 - \frac{1}{4}z^{-1}\right]\left[1 + \frac{1}{4}z^{-1}\right]}$$

8. Consider the following system:

$$H(z) = \frac{1 - 2z^{-1} + 2z^{-2} - z^{-3}}{(1 - z^{-1})(1 - 0.5z^{-1})(1 - 0.2z^{-1})}$$

a) Sketch the pole-zero pattern. Is this system stable?
b) Determine the impulse response of the system.

9. Determine and sketch the approximate magnitude response for the following filter: $H(z) = \frac{1-a}{1-az^{-1}}$ 0<a<1

10. A speech signal is sampled with a sampling period of 125ms. A frame of 256 samples is selected and a 256 point DFT is computed. What is the spacing between the DFT values in Hz.

11. Let x[n]={1, 2, 1, 0}. With N=4, compute the 4-point DFT.

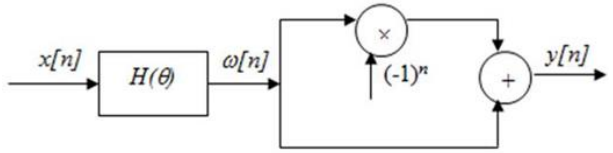
12. Find the 8-point DFT of $x[n]$ ={1, 1, 0, 0, 0, 0, 0, 0}, using conjugate symmetry property of $X[k]$.

Topic 1: Tutorial questions (Levels 2 & 3)

SOLO – Level 2

13. For the system in the figure below, sketch the output *y[n]* when the input *x[n]* is $\delta[n]$ and $H(\theta)$ is an ideal lowpass filter as follows:

$$H(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \frac{\pi}{2} \\ 0 & \frac{\pi}{2} < |\theta| \le \pi \end{cases}$$



14. A 3- second signal is sampled at f_s = 100 Hz. The maximum spectral spacing is to be $\Delta f = 0.25\ Hz$. How many samples are needed for the DFT and FFT ?

15. Classify the following system in terms of its linearity, time invariance, memory and causality: $y[n] = |x[-n]|$

SOLO – Level 3

1. If $y[n] = x[n] * h[n]$, then show that Y(z) = X(z) . H(z).

2. Find the z-transform, G(z), if $g[n] = \begin{cases} na^n & n \ge 0 \\ 0 & n < 0 \end{cases}$ where a is a constant

3. The frequency response of a band stop filter is given by

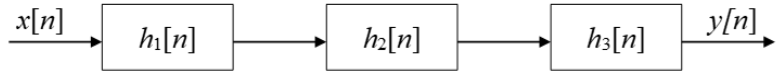
$$H_d(\theta) = \begin{cases} 1 & |\theta| \le \frac{\pi}{6} \\ 0 & \frac{\pi}{6} < |\theta| < \frac{\pi}{3} \\ 1 & \frac{\pi}{3} \le |\theta| \le \pi \end{cases}$$

Show that the impulse response $h_d[n]$ of the band stop filter is given by

$$h_d[n] = \delta[n] - \frac{\sin\left(\frac{\pi}{3}n\right)}{\pi n} + \frac{\sin\left(\frac{\pi}{6}n\right)}{\pi n}$$

SOLO – Level 3

4. Three systems with the following impulse responses are connected in cascade: $h_1[n] = \delta(n) - \delta(n-1)$; $h_2[n] = g[n]$; $h_3[n] = u[n]$; where $u[n]$ is the unit step sequence and $\delta[n]$ is the unit sample sequence (i.e an impulse).



What is the impulse response $h_c(n)$ of the overall system?

5. Show that the convolution of the two infinite duration sequences

$$p[n] = a^n u[n] \quad \text{and} \quad q[n] = b^n u[n]$$

for all *n*, where *u*[*n*] is the unit step function and $a \neq b$, is given by

$$y[n] = \frac{a^{n+1} - b^{n+1}}{a - b}$$

6. Consider a system with an input x[n] and an output y[n] which satisfies the following difference equation:

$$y[n] = n\, y[n-1] + x[n]$$

If $x[n] = \delta[n]$, determine $y[n]$ for all values of n.

Topic 1: Tutorial questions (Level 4)

SOLO – Level 4

1. A system L is known to be linear. The responses of L to inputs $x_1[n]$ and $x_2[n]$ are given by $y_1[n]$ and $y_2[n]$ respectively (see Figure 1).

   (a) Determine whether the system L could be time-invariant.

   (b) If the input x[n] to the system is $\delta[n]$, what is the system response y[n]?

2. Consider the interconnection of LTI systems as shown in Figure 2:

   (a) Express the overall impulse response in terms of $h_1[n]$, $h_2[n]$, $h_3[n]$ and $h_4[n]$.

   (b) Determine *h[n]* when

   $h_1[n] = \{\frac{1}{2}\ \underset{\uparrow}{\frac{1}{4}}\ \frac{1}{2}\}$; $h_2[n] = h_3[n] = (n+1)\,u[n]$; $h_4[n] = \delta[n-2]$

   (c) Determine the response of the system in part(b) if

   $x[n] = \delta[n+2] + 3\delta[n-1] - 4\delta[n-3]$

3. Find a difference equation to implement a system that has a unit impulse response:

   $$h[n] = (\frac{1}{4})^n \cos(n\frac{\pi}{3})\, u[n]$$

4. An Lth order moving average filter is a linear time invariant system that, for an input $x[n]$, produces the output $y[n] = \frac{1}{L+1}\sum_{k=0}^{L} x[n-k]$. Show that the frequency response of this system is $H(\theta) = \frac{1}{L+1} e^{-jL\theta/2} \frac{\sin[(L+1)\theta/2]}{\sin(\frac{\theta}{2})}$

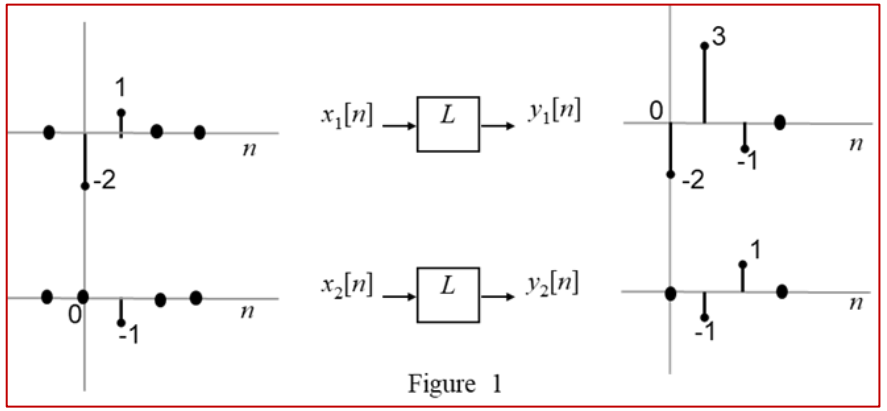


Figure 1

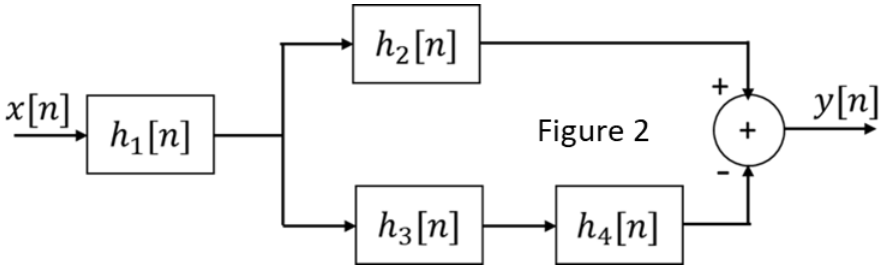


Figure 2

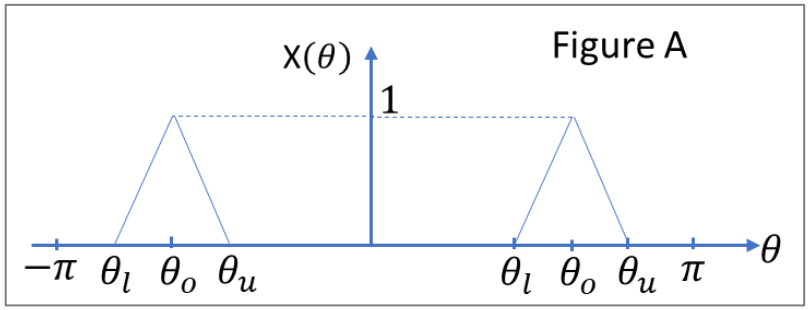
Topic 1: Tutorial questions (Level 5)

SOLO – Level 5

1. Consider a bandpass filter with a spectrum X(θ) as shown in Figure A. Show that x[n] can be formulated as:

$$x[n] = (k_p + k_q)\cos(n\theta_0) - k_p\cos(n\theta_u) - k_q\cos(n\theta_l)$$

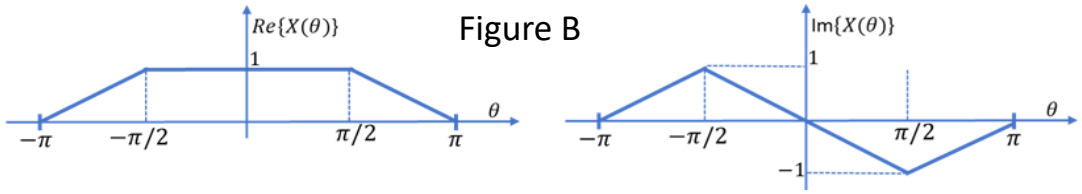
$$k_p = \frac{1}{\pi n^2(\theta_u - \theta_0)}; \qquad k_q = \frac{1}{\pi n^2(\theta_0 - \theta_l)}; \qquad \theta = 2\pi\frac{f}{f_s}$$



2. The centre of gravity of a sequence is defined by $c = \frac{\sum_{n=-\infty}^{\infty} n\,x[n]}{\sum_{n=-\infty}^{\infty} x[n]}$ and is used as a measure of the *time delay* of a sequence: Find (using the properties of DTFT)

   (a) an expression for *c* in terms of the DTFT of x[n].

   (b) the value of *c* for the sequence x[n] that has a DTFT as shown in Figure B.



3. For the sequence x[n] plotted in Figure C, evaluate the integral $\int_{-\pi}^{\pi}\left|\frac{d}{d\theta}X(\theta)\right|^2 d\theta$.

